

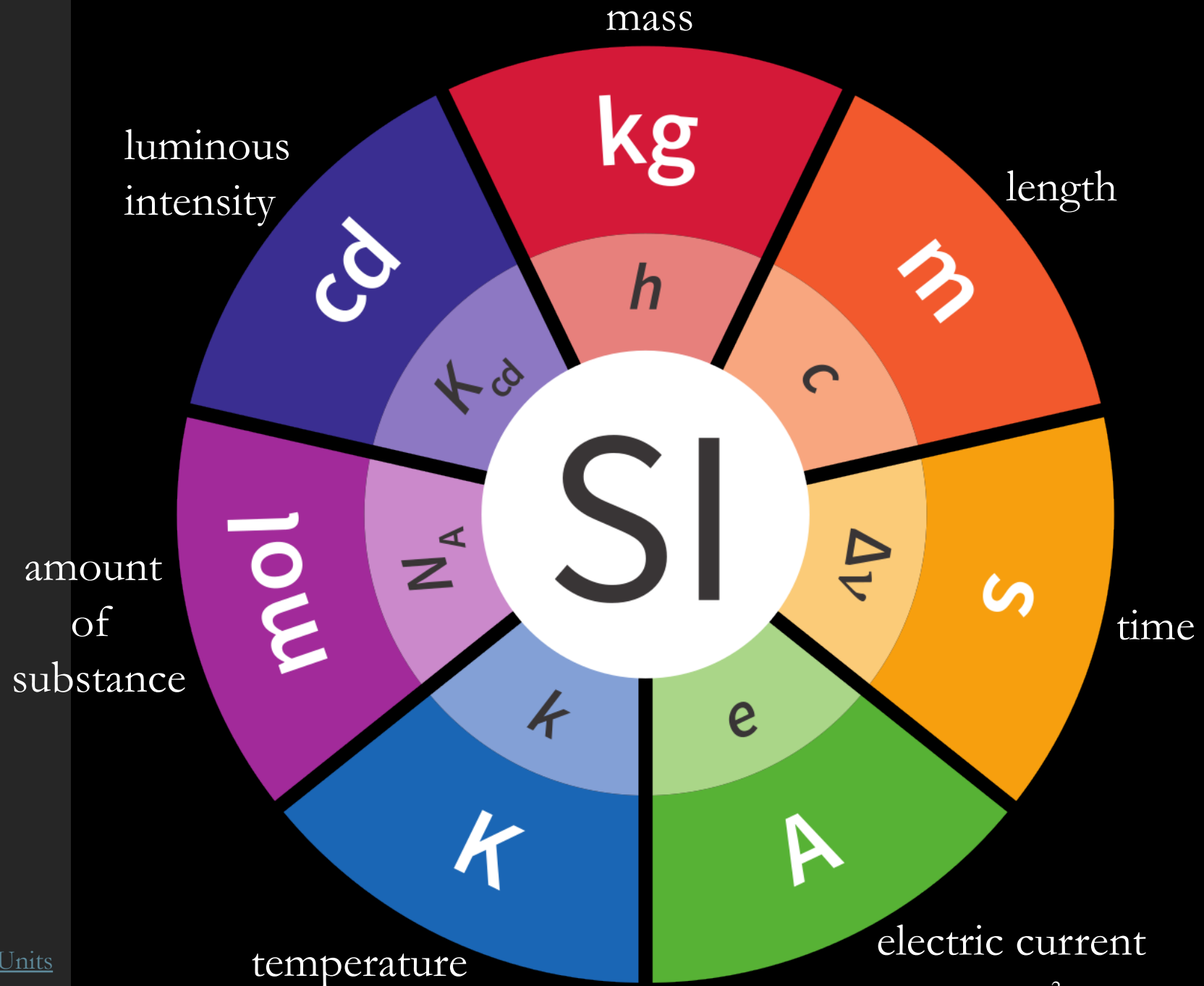
CY1308 Physics



LECTURE 1

SI UNITS, ON MODELS, KINEMATICS (Y&F CHAP 1,2,3)

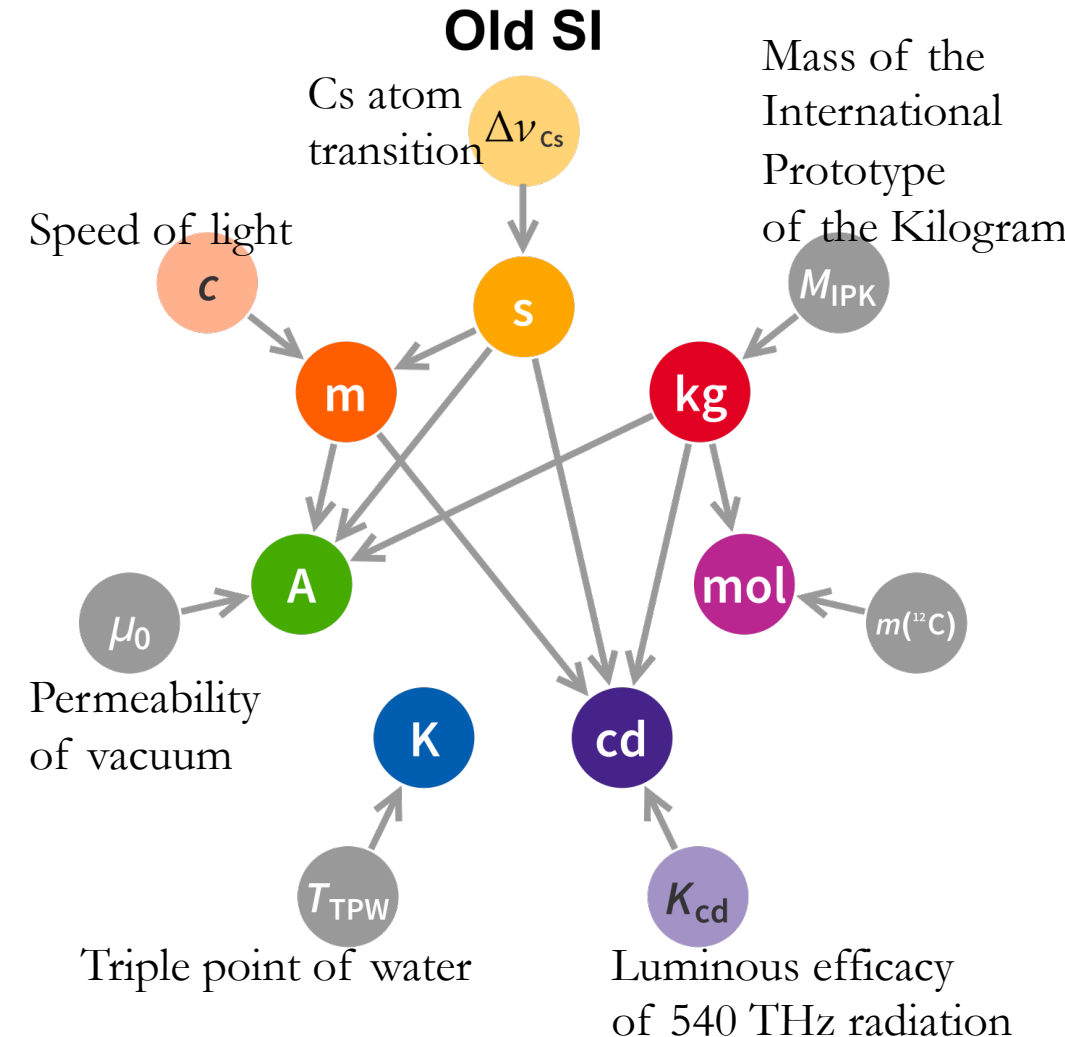
SI Units



Système Internationale (SI) d'unités: Original conception

Original conception (France, 1799):

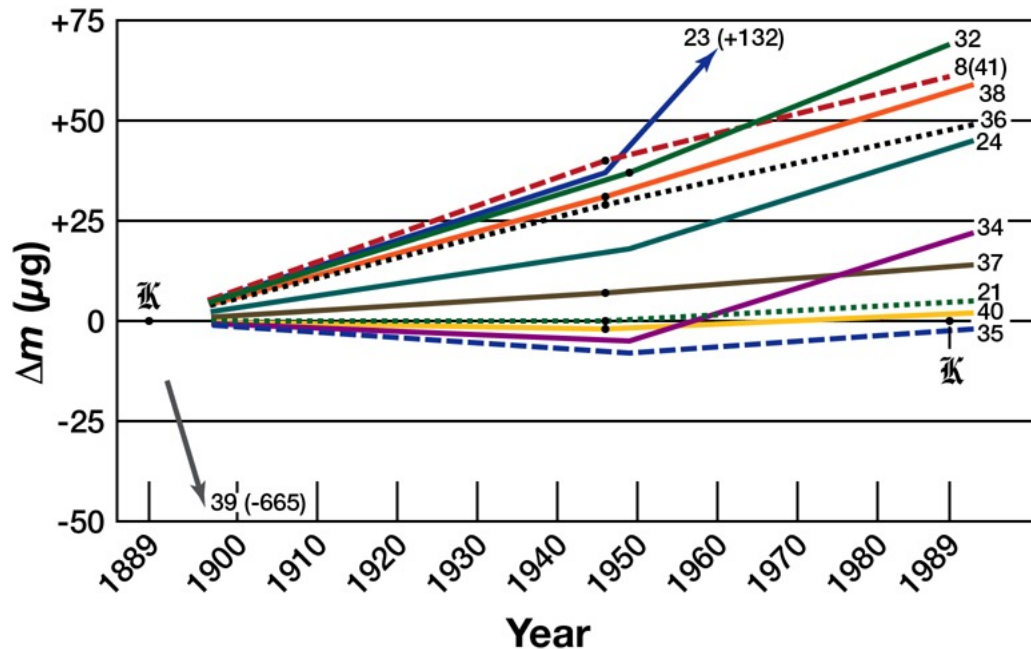
- The metric system was originally conceived as a system of measurement that was derivable from unchanging phenomena.
- Practical limitations necessitated the use of artefacts – the prototype metre and prototype kilogram. Some were difficult to precisely realise in a laboratory, such as the kelvin, which was defined in terms of the triple point of water.
- Masses of the prototype kilogram and its secondary copies have shown small variations relative to each other over time.
- As such, they are not thought to be adequate for the increasing accuracy demanded by science, prompting a search for a suitable replacement.



Old SI: Mass drift problem

Mass drift of prototype kilogram and derivatives:

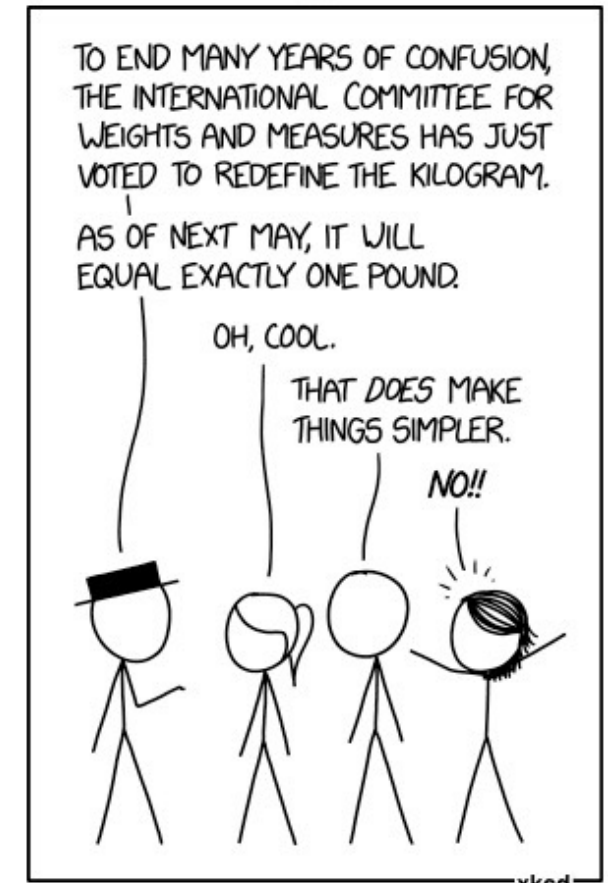
- Masses of the prototype kilogram and its secondary copies have shown small variations relative to each other over time, prompting a search for a suitable replacement.



Mass drift over time of national prototypes K21–K40, plus two of the international prototype's sister copies: K32 and K8(41). All mass changes are relative to the IPK.

Source: Wikipedia,
https://en.wikipedia.org/wiki/2019_redefinition_of_the_SI_base_units

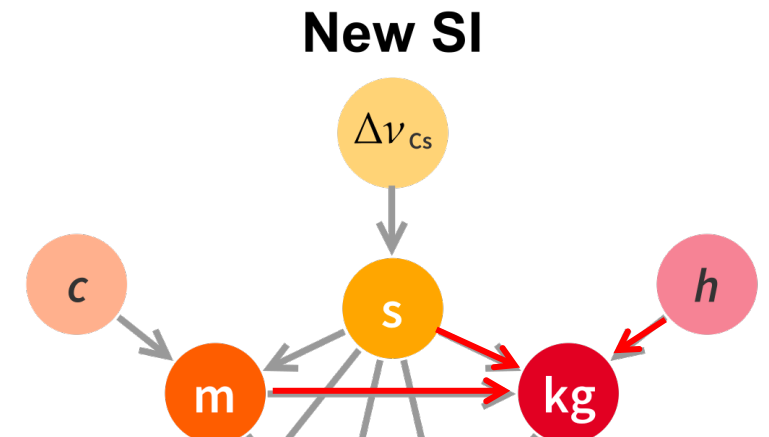
Kilogram



Mass and its relation to fundamental constants

- Before 2019, Planck's constant h had some uncertainty because it is determined experimentally. Frequency f (or the SI unit for time) and speed of light c were already exact by 2019.
- In contrast, the IPK mass was absolutely certain because by definition it is 1 kg.
- But this kilogram artefact (IPK) is practically inaccessible (being sealed in a vacuum jar and stored in a basement vault in Paris), and there are mass drifts with its derivative artefacts.
- Solution: define Planck's constant to be exact, and the mass of the kilogram will follow. The IPK will no longer be exactly 1 kg.

$$E = hf = mc^2$$
$$\frac{hf}{c^2} = m$$



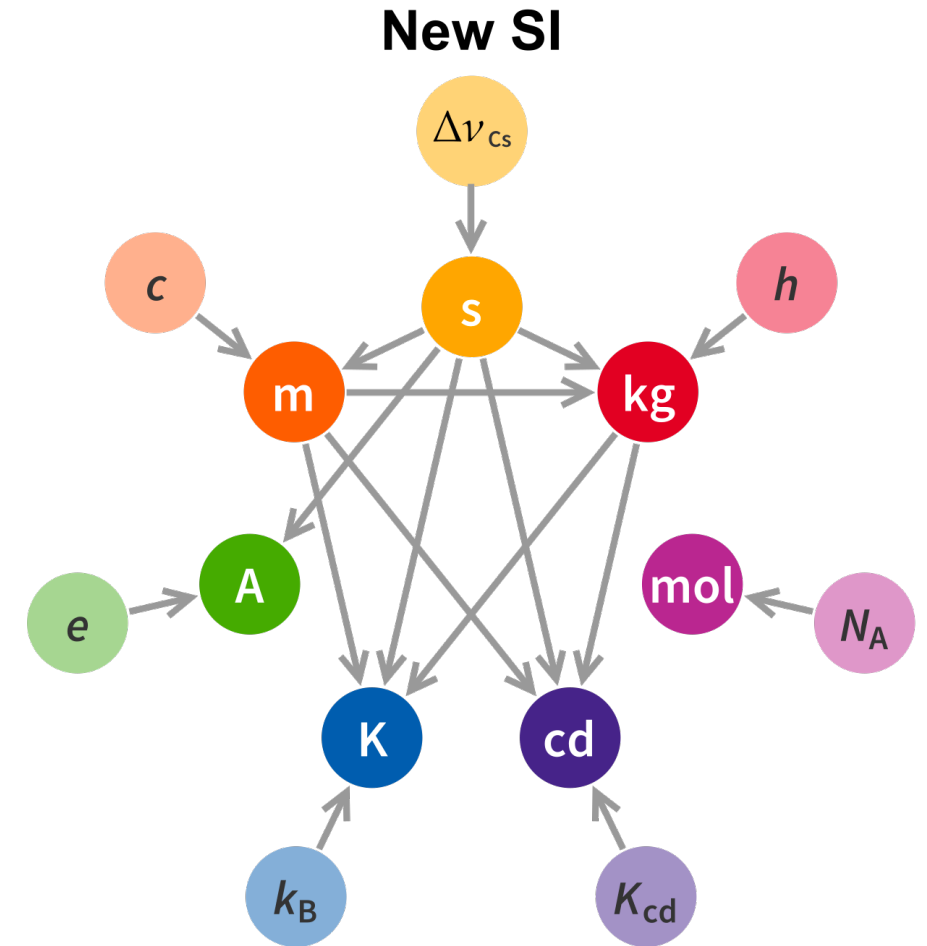
Redefinition of SI (2019)

Precedence for change

- **1960:** redefined the SI metre in terms of the wavelength of krypton-86 radiation, discarded the pre-SI metre bar (artefact).
- **1967:** redefined original definition of the second, which was based on Earth's average rotation from 1750 to 1892, with a definition based on the frequency of the radiation emitted or absorbed with a caesium-133 atomic transition between two hyperfine levels of the ground state.
- **1983:** redefined the 1960 definition of the metre with one based on the second, by giving an exact definition of the speed of light in units of metres per second.

Redefinition of SI (2019)

- Fundamental change to the definition of SI units. SI units became *entirely* derivable from natural phenomena (no artefacts). Most units based on fundamental physical constants.



Source: Wikipedia

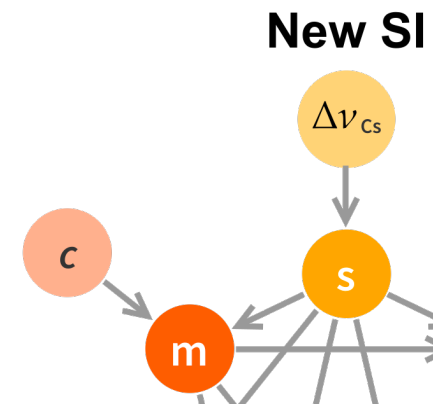
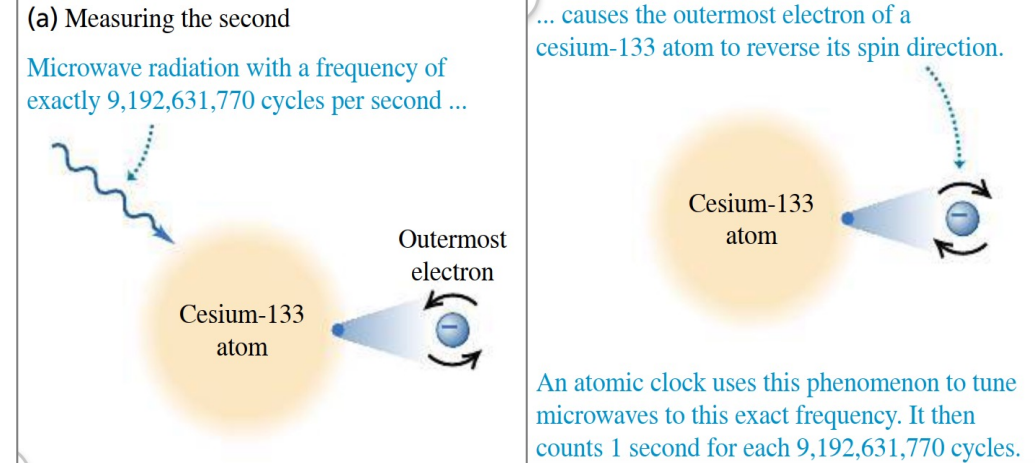
https://en.wikipedia.org/wiki/2019_redefinition_of_the_SI_base_units

Redefinition of SI (2019): Fundamental constants “fixed”

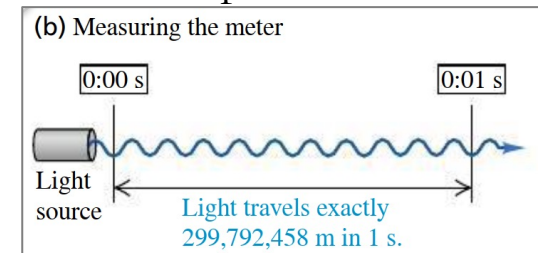
Defining the constants (as opposed to determining them by experiment)

- International Bureau for Weights and Measures proposed that 4 further constants of nature (first 4 below) be defined to have exact values.
- Planck constant h** is exactly $6.62607015 \times 10^{-34}$ joule-second (J·s).
- Elementary charge e** is exactly $1.602176634 \times 10^{-19}$ coulomb (C).
- Boltzmann constant k** is exactly 1.380649×10^{-23} joule per kelvin (J·K⁻¹).
- Avogadro constant N_A** is exactly $6.02214076 \times 10^{23}$ reciprocal mole (mol⁻¹).
- Speed of light c** is exactly 299792458 metres per second (m·s⁻¹).
- Ground state hyperfine structure transition frequency** of caesium-133 atom $\Delta\nu_{\text{Cs}}$ is exactly 9192631770 hertz (Hz).
- Luminous efficacy K_{cd}** of monochromatic radiation of frequency 540×10^{12} Hz (540 THz) – a frequency of green-colored light at approximately the peak sensitivity of the human eye – is exactly 683 lumens per watt (lm·W⁻¹).

Time-keeper: caesium-133 atomic clock



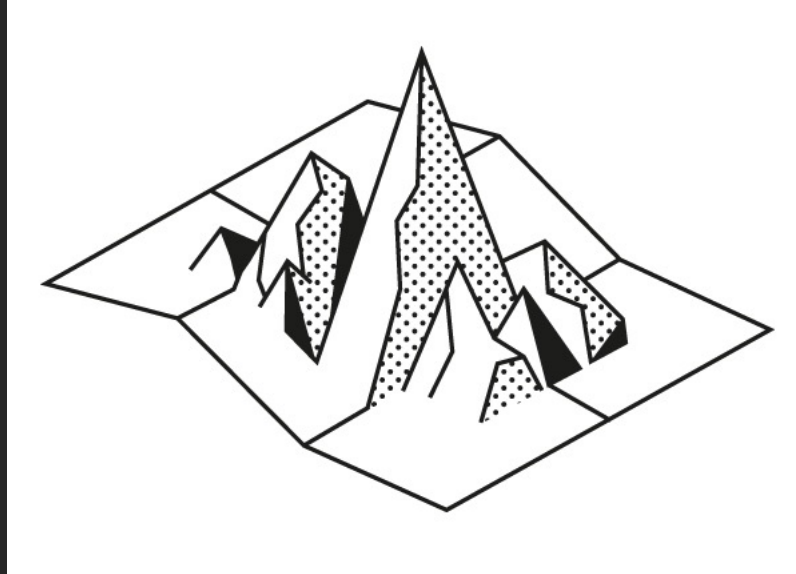
Length from the universal speed limit c



Source: Young and Freedman

Models in Physics

Models as maps to the
terrain of the physical world



Models in physics as maps of the terrain (physical world)

Consider the legend of Galileo's experiments on falling objects. He is supposed to have dropped them from the Leaning Tower of Pisa in Italy.

Galileo's pupil, Vincenzo Viviani, wrote in 1717 in *Racconto storico della vita di Galileo Galilei*, p. 606: [...dimostrando ciò con replicate esperienze, fatte dall'altezza del Campanile di Pisa con l'intervento delli altri lettori e filosofi e di tutta la scolaresca... ***[...Galileo showed this [all bodies, whatever their weights, fall with equal speeds] by repeated experiments made from the height of the Leaning Tower of Pisa in the presence of other professors and all the students...]***].

If Galileo had compared the motion of a rock and a feather, but had not “idealized” the motion, by perhaps imagining that the air could slow down the feather more than the rock, he would not have made the conclusion.

It is through an idealization or model of reality – by creating a map of the terrain – where air resistance is absent or equal for both objects, that Galileo could construct a model that captured the essence of the motion without all the complications of reality.

According to legend, Galileo investigated falling objects by dropping them from the Leaning Tower of Pisa, Italy, ...



... and he studied pendulum motion by observing the swinging chandelier in the adjacent cathedral.

Image source: Young & Freedman

Physics is an empirical science

Physics, like chemistry and biology, is an *empirical* science.

It contains propositions in the theories which are not deducible from definitions and must be tested with **empirical** evidence from observations about nature. At the same time, physics is a *dialogue* with nature and observations can lead to new theory, and vice versa.

“The experience of subjecting a theoretical conjecture to an experimental test is humbling. If the conjecture turns out to be wrong, it must be adjusted. Becoming a physicist brings with it the privilege of retaining your childhood curiosity throughout your adult life. There is no need to pretend you know more than you actually do, and you can admit mistakes if proven wrong by experience, just like a child who is seeking to learn about the world. Doing pure theory without worrying about experimental verification actually deprives one from the pleasure of learning something new about nature.”

Quote from Scientific American, “Theoretical Physics Is Pointless without Experimental Tests”:
<https://blogs.scientificamerican.com/observations/theoretical-physics-is-pointless-without-experimental-tests/>

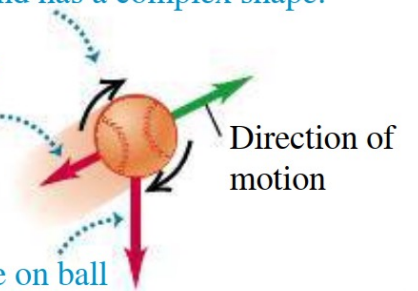
Figure 1.2 To simplify the analysis of (a) a baseball in flight, we use (b) an idealized model.

(a) A real baseball in flight

A baseball spins and has a complex shape.

Air resistance and wind exert forces on the ball.

Gravitational force on ball depends on altitude.



(b) An idealized model of the baseball

Treat the baseball as a point object (particle).

No air resistance.

Gravitational force on ball is constant.

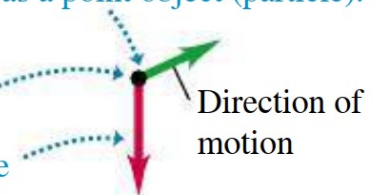
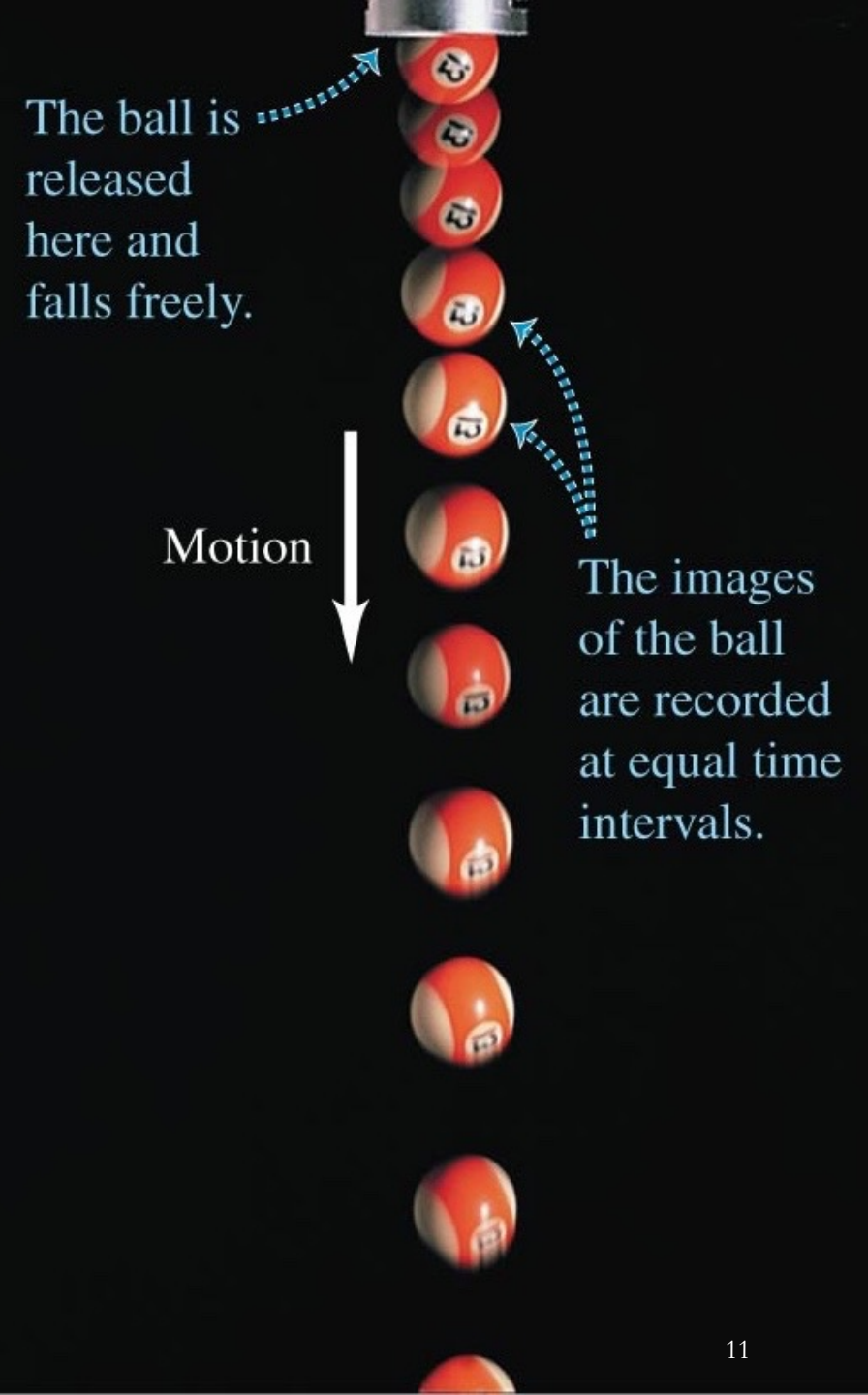


Image source: Young & Freedman

Kinematics: Describing motion

From 1 to 2 and 3 dimensions

Image source: Young & Freedman



Position, displacement and distance

Position and coordinate system

- To study motion of an object, we need to specify its **position** in a coordinate system.
- In 3D space, the coordinates will be a trio of numbers (x, y, z) for the Cartesian system. The position-time graph on the right is for an object moving in 1D.

Displacement (vector quantity)

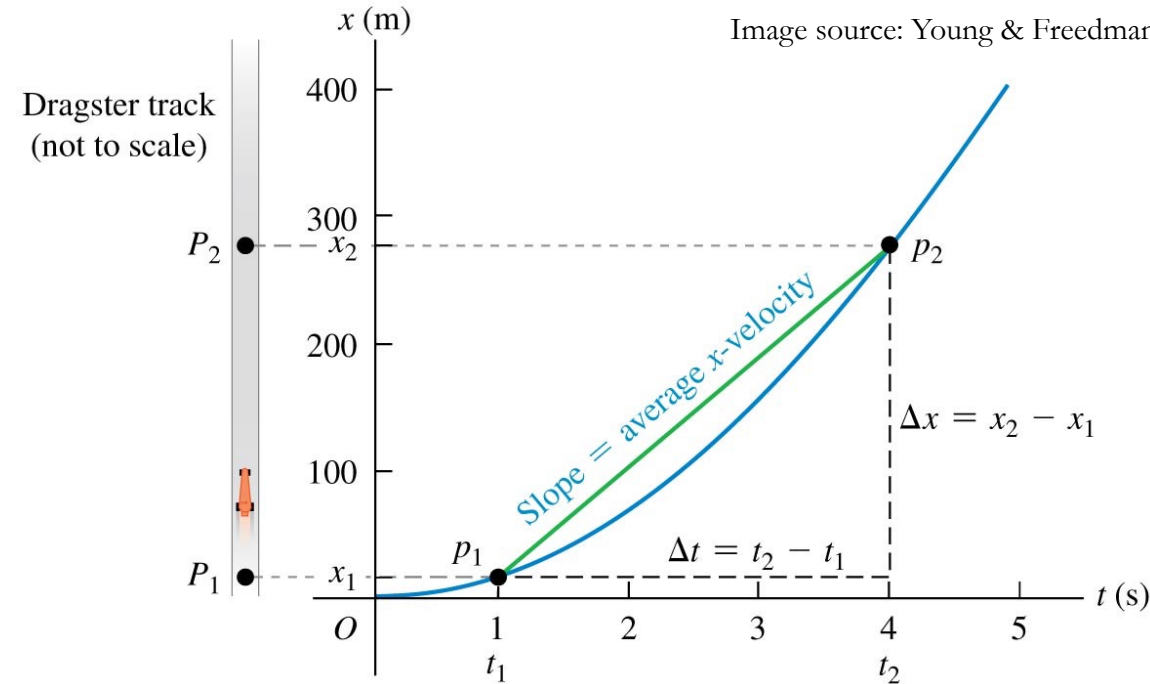
- Displacement is the change in position of a body (over a time interval): $\Delta \vec{x} = \vec{x}_f - \vec{x}_i$
- Only concerned with **end points**.

Distance (scalar quantity)

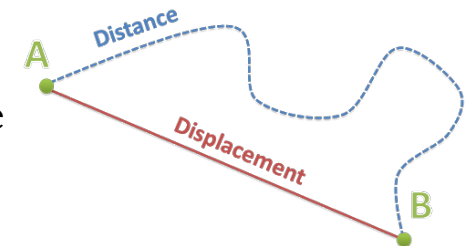
- Understood to be the length of an **entire path** taken.

Note that change in any physical quantity is always its *final* value minus its *initial* value.

Position-time graph



Displacement vs distance



Velocity and speed (SI units: m s^{-1})

Average velocity (vector quantity)

- Change in a particle's position $\Delta \vec{x}$ over a **finite** time interval Δt
- Average velocity: $\vec{v}_{av} = \frac{\Delta \vec{x}}{\Delta t}$

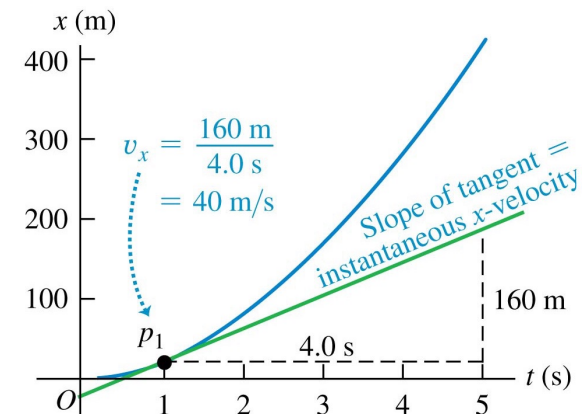
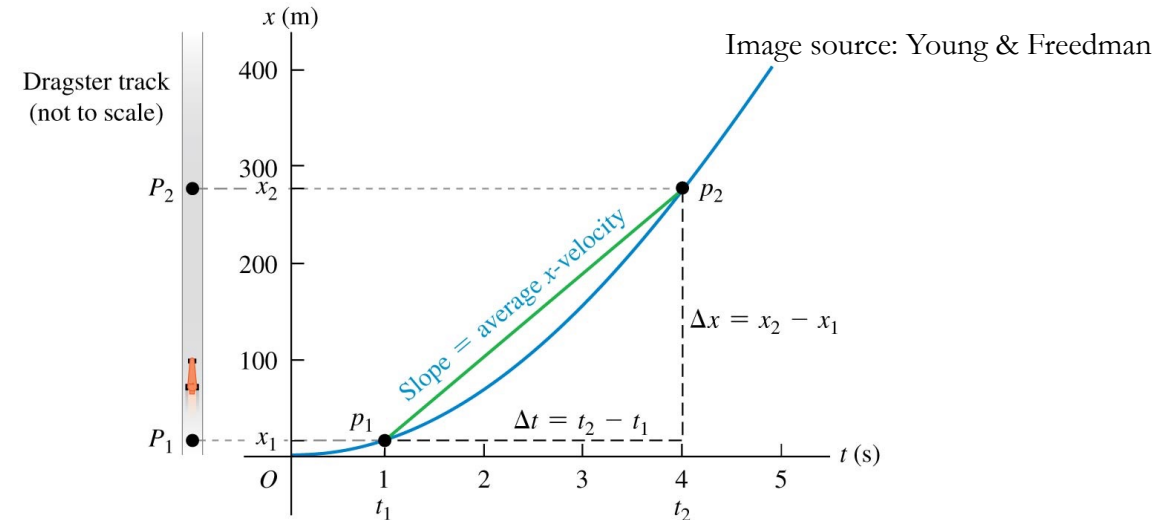
Instantaneous velocity (vector quantity)

- In the limit that the time interval becomes infinitesimally small: $\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{x}}{\Delta t} = \frac{d\vec{x}}{dt}$

Speed (scalar quantity)

- Defined as the magnitude of velocity.

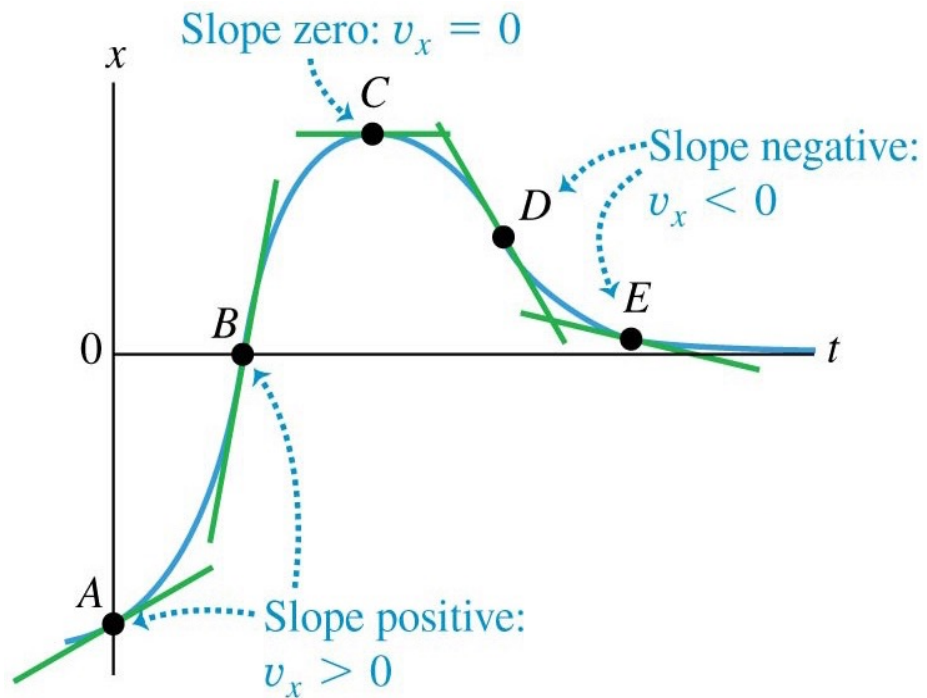
Note that (instantaneous) velocity is often defined in GCE A level texts to be the (time) rate of change of *displacement*. But it should not be. It is the rate of change of *position*.



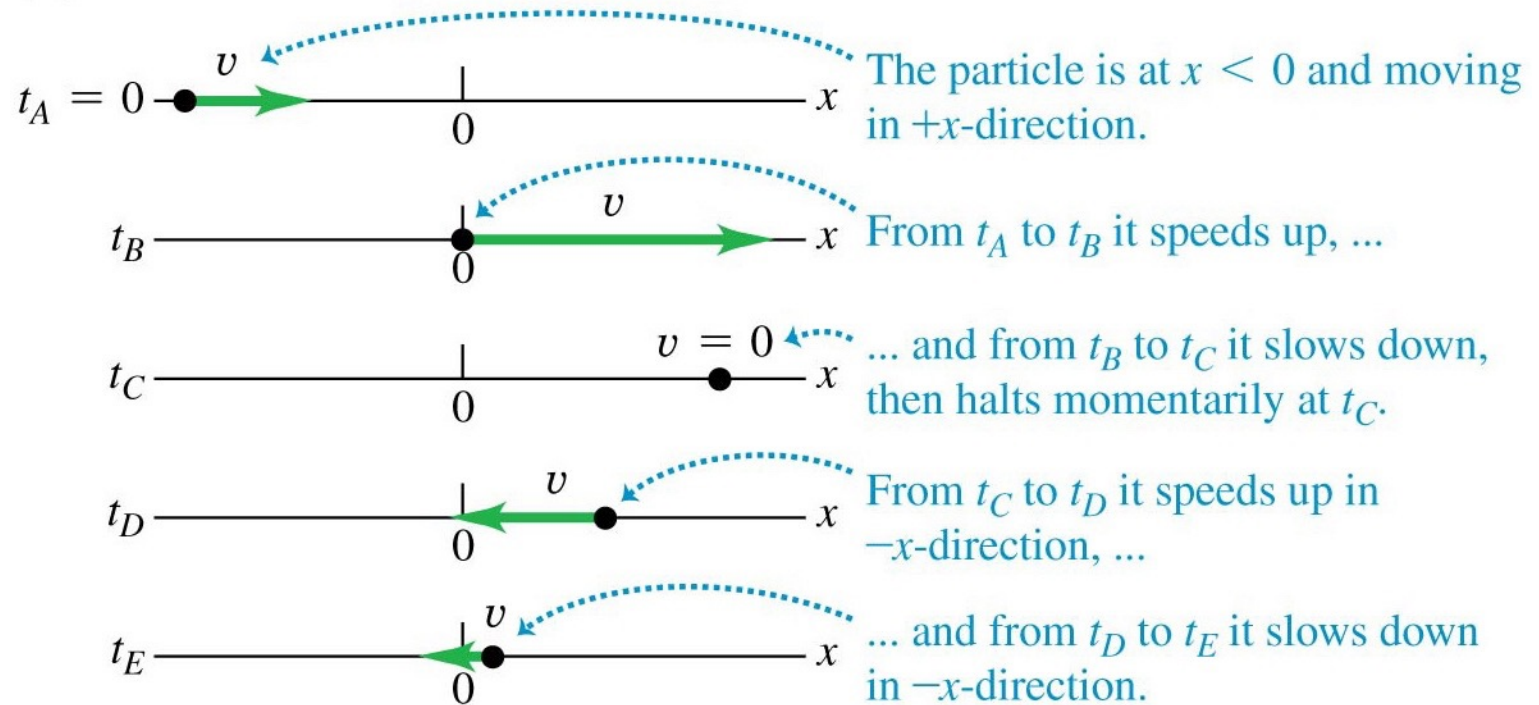
The instantaneous x -velocity v_x at any given point equals the slope of the tangent to the x - t curve at that point.

Gradient of Position-time graph is (instantaneous) velocity

Position-time graph of particle



Motion diagram of particle



Acceleration (SI units: m s⁻²)

Average acceleration (vector quantity)

- The change in velocity over a finite time interval is the average acceleration:
$$\vec{a}_{av} = \frac{\Delta \vec{v}}{\Delta t}$$

Instantaneous acceleration (vector quantity)

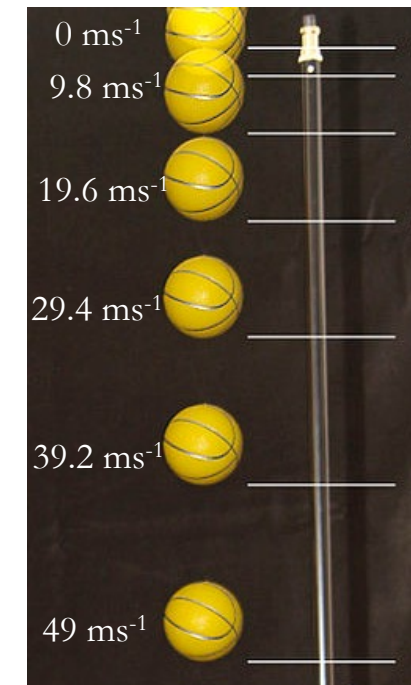
- In the limit that the time interval becomes infinitesimally small, the instantaneous acceleration is:

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2 \vec{x}}{dt^2}$$

The units of acceleration give a sense of how to understand this quantity. It is given by metres per second per second, i.e. it is the change in velocity (m s⁻¹) every second.

Falling ball : multiple images taken at constant 1s intervals

where acceleration is constant at 9.8 m/s/s



Source

https://commons.wikimedia.org/wiki/File:Falling_ball_original_plus_text.jpg

Acceleration (SI units: m s^{-2})

Average acceleration (vector quantity)

- The change in velocity over a finite time interval is the average acceleration:
$$\vec{a}_{av} = \frac{\Delta \vec{v}}{\Delta t}$$

Instantaneous acceleration (vector quantity)

- In the limit that the time interval becomes infinitesimally small, the instantaneous acceleration is:

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2 \vec{x}}{dt^2}$$

The gradient of a velocity-time graph is the acceleration.

Velocity-time graph

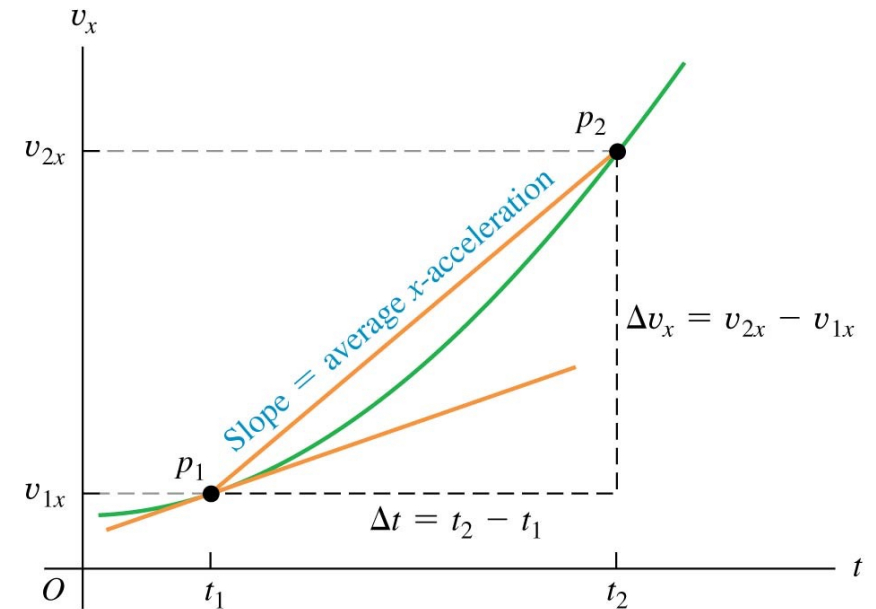


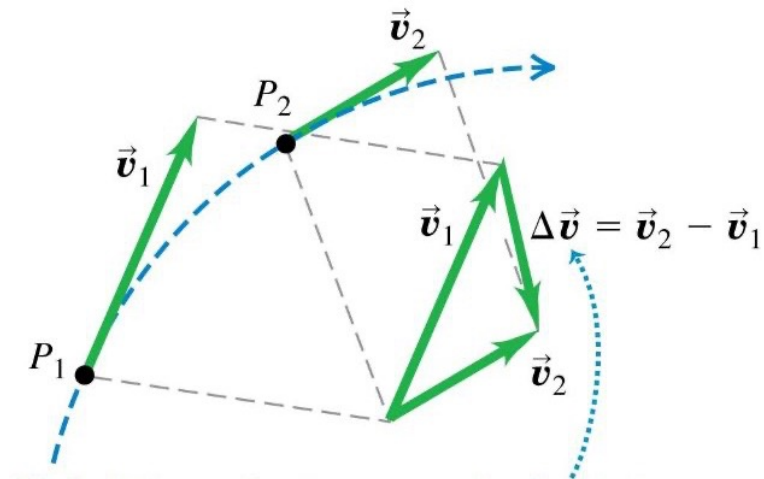
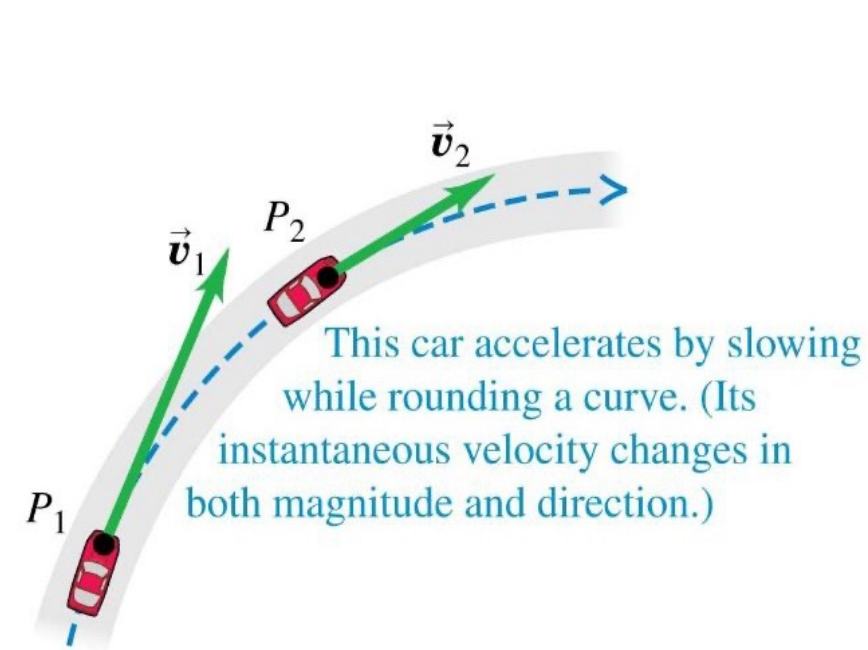
Image source: Young & Freedman

Acceleration (SI units: m s^{-2})

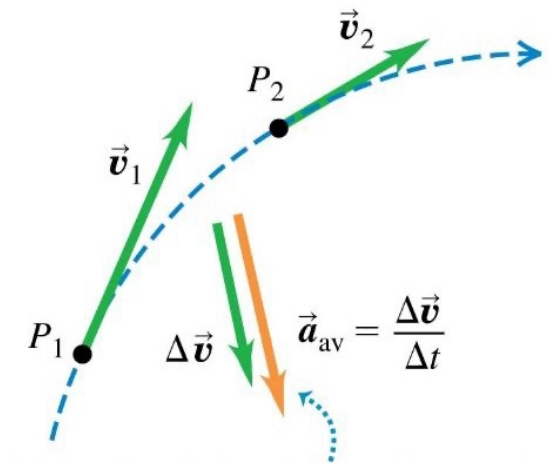
Important note:

Even if speed is constant, as long as the direction of the object changes, the velocity is changing. This means that a car has a non-zero acceleration if it rounds a bend at constant speed (because its direction changes hence its velocity is not constant).

Example: Finding average acceleration in 2D motion with a vector diagram.
$$\vec{a}_{av} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1}$$



To find the car's average acceleration between P_1 and P_2 , we first find the change in velocity $\Delta \vec{v}$ by subtracting $\vec{v}_1$ from $\vec{v}_2$. (Notice that $\vec{v}_1 + \Delta \vec{v} = \vec{v}_2$.)

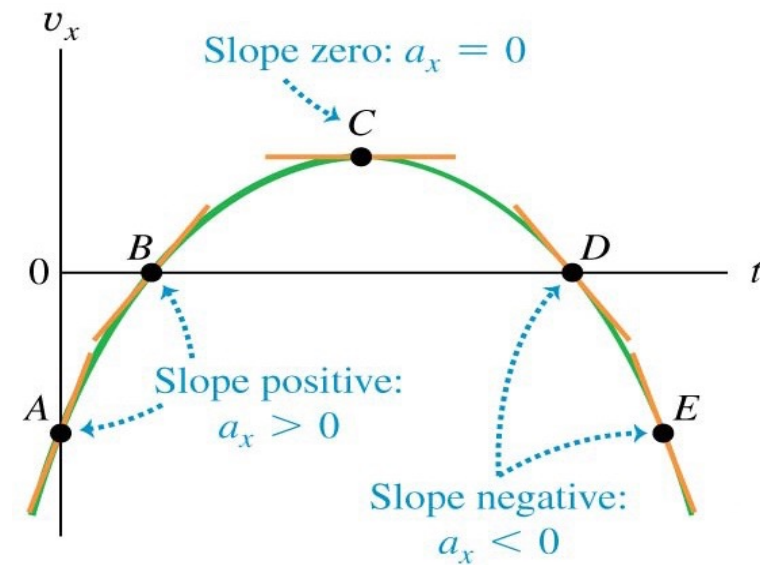


The average acceleration has the same direction as the change in velocity, $\Delta \vec{v}$.

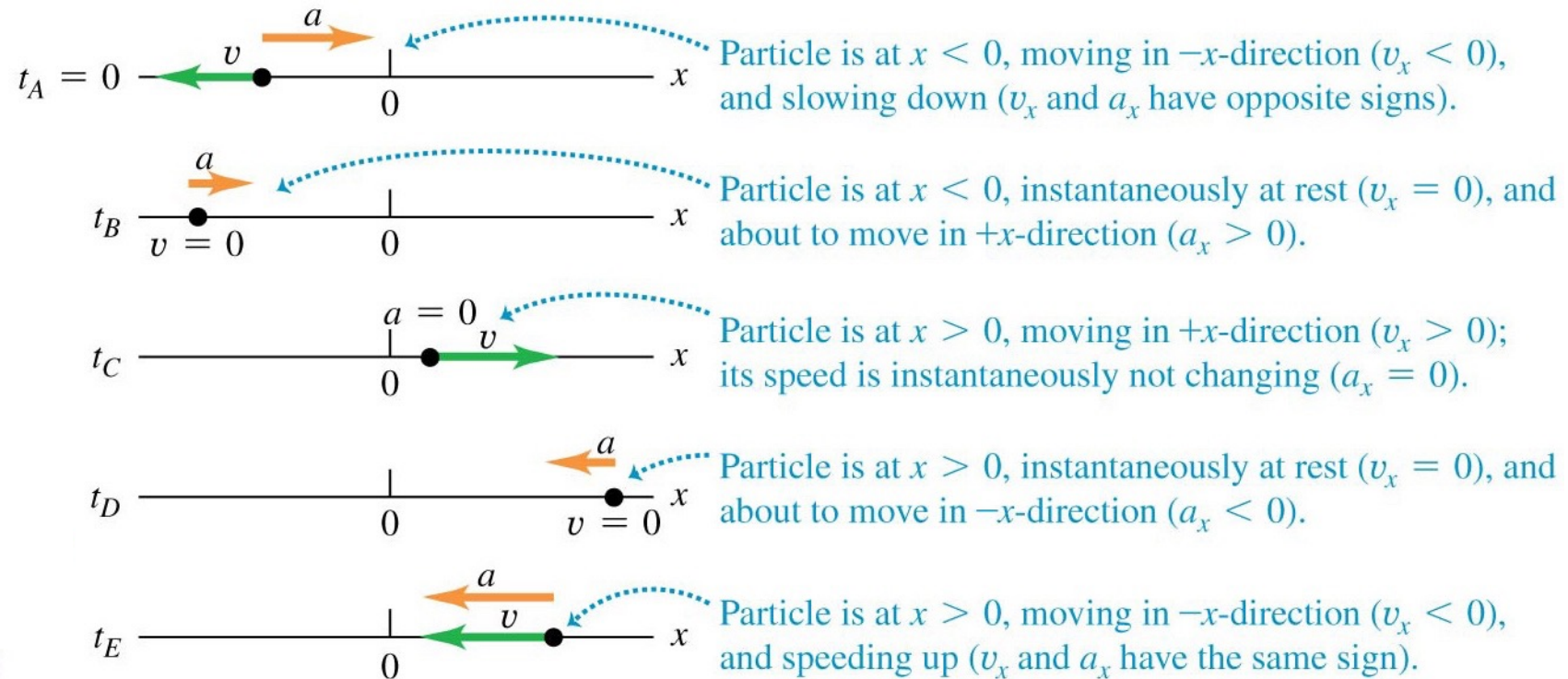
Image source: Young & Freedman

Gradient of velocity-time graph is (instantaneous) acceleration

Velocity-time graph of particle



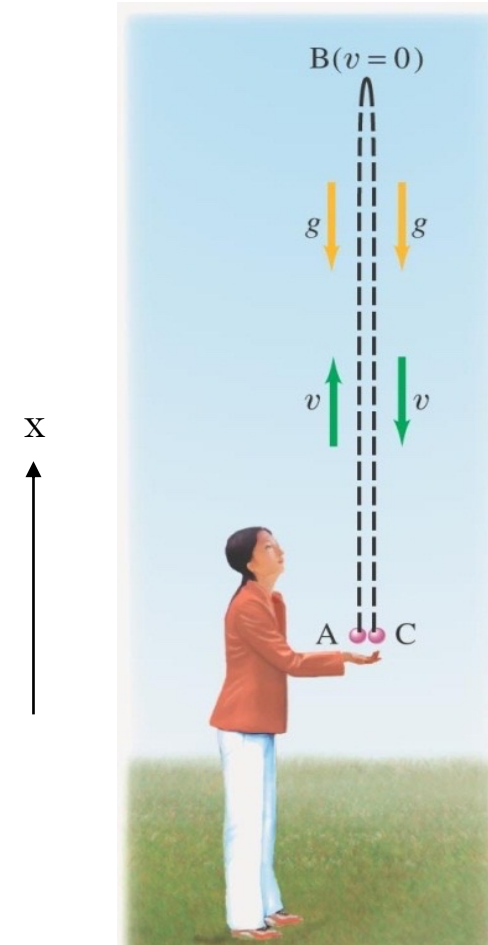
Motion diagram of particle



Sign of acceleration and velocity and motion

Important: do not use the layman understanding of acceleration to mean speeding up. When there is an (non-zero) acceleration, it just means that velocity changes.

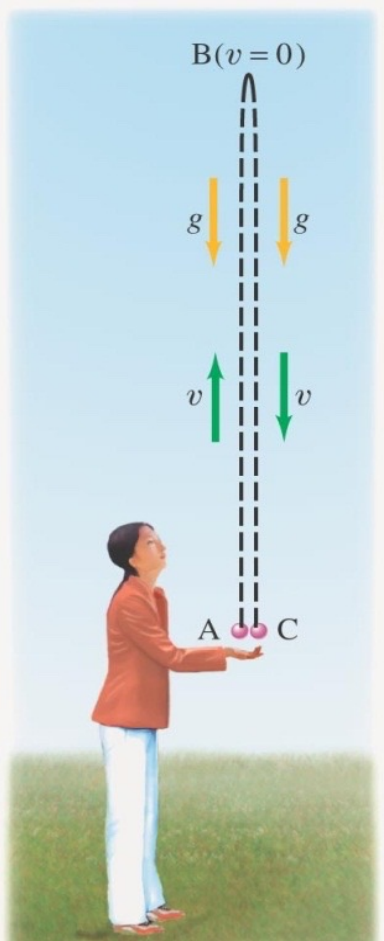
If x -velocity is:	... x -acceleration is:
Positive & increasing (getting more positive)	Positive: Particle is moving in $+x$ -direction & speeding up
Positive & decreasing (getting less positive)	Negative: Particle is moving in $+x$ -direction & slowing down
Negative & increasing (getting less negative)	Positive: Particle is moving in $-x$ -direction & slowing down
Negative & decreasing (getting more negative)	Negative: Particle is moving in $-x$ -direction & speeding up



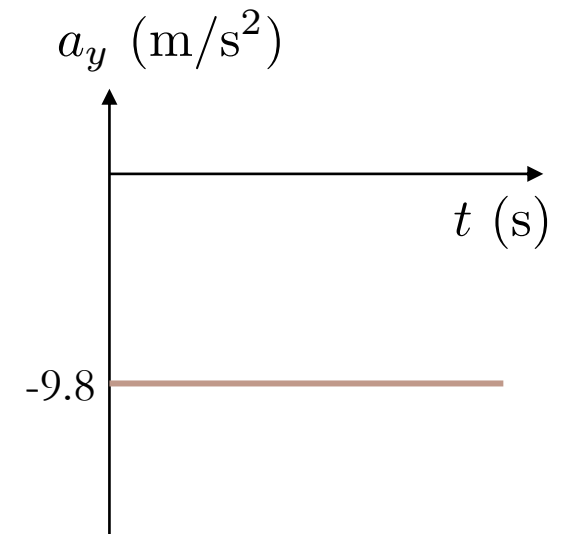
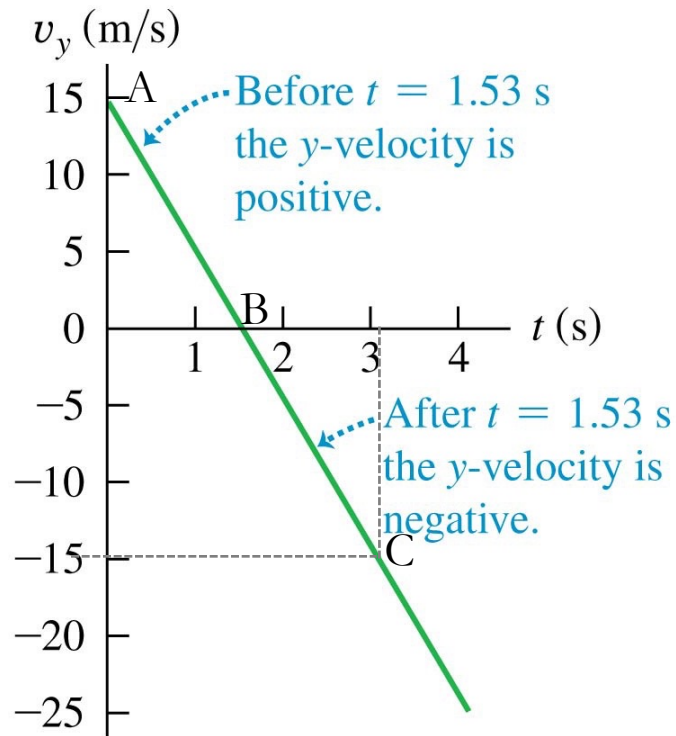
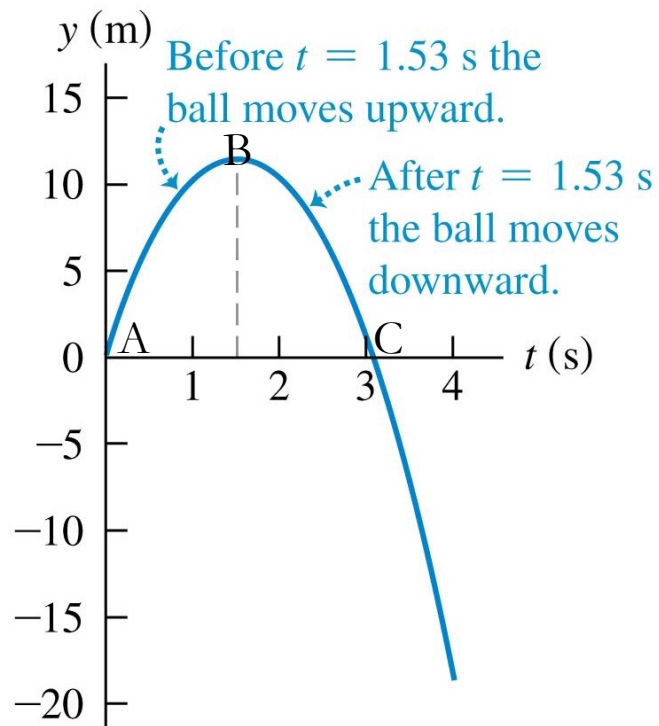
Constant acceleration g is in the $-x$ direction

Motion described with acceleration, velocity and position

A man throws a ball up.

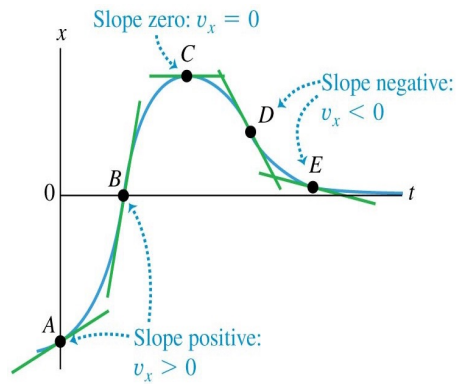


- The coordinate system (arbitrary) is such that upwards is the $+y$ direction.
- The ball's initial velocity is 15 m/s, and it reaches a maximum height after 1.53 s and back to the hand in twice that time, i.e. $2 \times 1.53 \text{ s} = 3.06 \text{ s}$.
- The earth's gravitational pull produces a constant acceleration of magnitude 9.8 m/s^2 pointing in the $-y$ direction.



Quick summary: General relations in Kinematics

Gradient of a position-time graph is velocity.

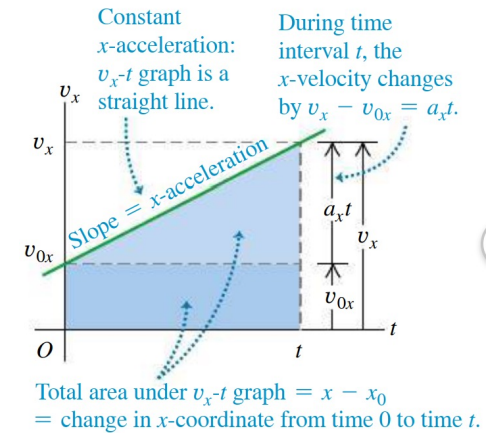


$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{x}}{\Delta t} = \frac{d\vec{x}}{dt}$$

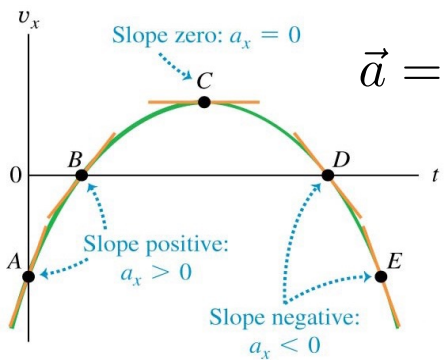
$$\Rightarrow \int_{\vec{x}_i}^{\vec{x}_f} d\vec{x} = \int_{t_i}^{t_f} \vec{v}(t) dt$$

$$\vec{x}_f - \vec{x}_i = \int_{t_i}^{t_f} \vec{v}(t) dt$$

Area under a velocity-time graph is displacement (change in position).



Gradient of a velocity-time graph is acceleration.

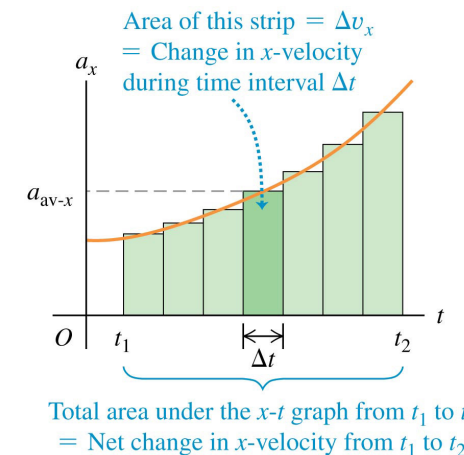


$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2 \vec{x}}{dt^2}$$

$$\Rightarrow \int_{\vec{v}_i}^{\vec{v}_f} d\vec{v} = \int_{t_i}^{t_f} \vec{a}(t) dt$$

$$\vec{v}_f - \vec{v}_i = \int_{t_i}^{t_f} \vec{a}(t) dt$$

Area under an acceleration-time graph is change in velocity.



Case of constant acceleration: deriving equations of motion (1)

Considering the simpler case of 1D motion (drop the vector sign), we perform the integrals on the previous slide.

If initial time is set to be zero, and the final time is a variable, i.e. $t_i = 0, t_f = t$

$$v_f - v_i = \int_0^t a \, dt$$

$$v_f - v_i = at$$

Understanding the initial velocity is a constant and the final velocity is a function of time, we can change notation to:

$$v(t) = v_0 + at$$

From the previous slide, we can also derive: $x_f - x_i = \int_0^t v(t) \, dt = \int_0^t (v_0 + at) \, dt$

$$x_f - x_i = v_0 t + \frac{1}{2} at^2$$

Again, a change of notation gives:

$$\underbrace{x(t) - x_0}_{\text{displacement}} = v_0 t + \frac{1}{2} at^2$$

displacement

Case of constant acceleration: deriving equations of motion (2)

We can arrive at a final, independent equation of motion by using a chain-rule trick:

$$a = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt}$$

$$a = v \frac{dv}{dx}$$

$$\int_{x_i}^{x_f} a \, dx = \int_{v_i}^{v_f} v \, dv$$

$$a \int_{x_i}^{x_f} dx = \int_{v_i}^{v_f} v \, dv$$

$$a(x_f - x_i) = \frac{1}{2}(v_f^2 - v_i^2)$$

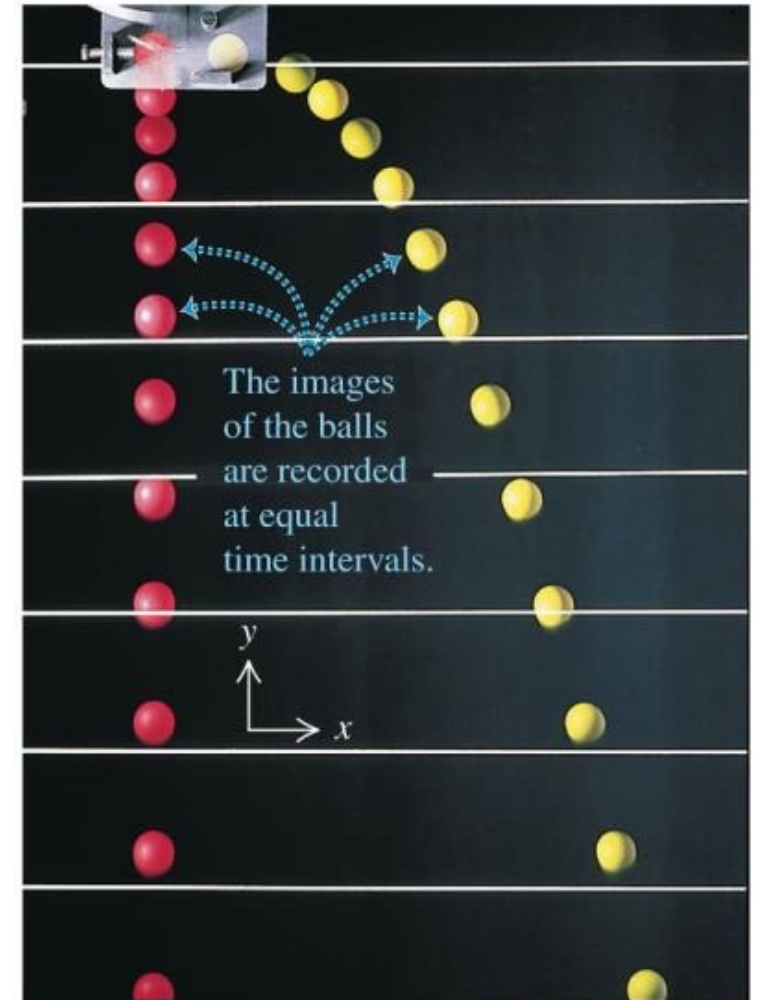
Rewriting with the same change of notation, we have:

$$v^2 = v_0^2 + 2a \underbrace{(x - x_0)}_{\text{displacement}}$$

displacement

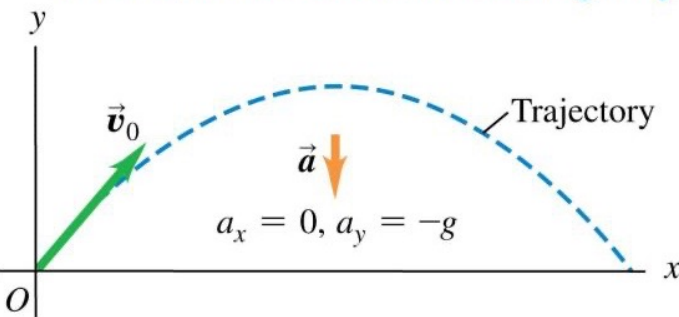
Perpendicular directions are independent or separable

- The red ball is dropped at the same time that the yellow ball is fired horizontally.
- The strobe marks equal time intervals.
- We can analyze projectile motion as horizontal motion with constant velocity and vertical motion with constant acceleration: $a_x = 0$ and $a_y = -g$.
- Recommended video: Paul Hewitt's conceptual physics <https://youtu.be/qErh402eJgI>

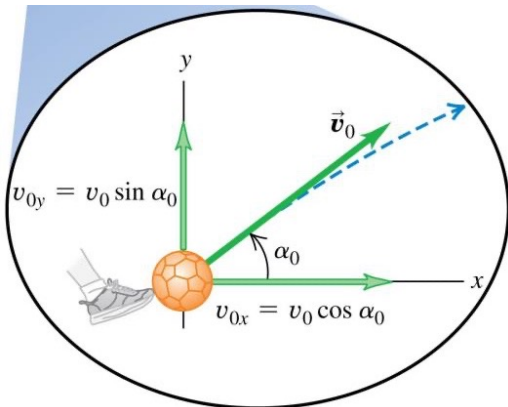


Projectile motion (2D)

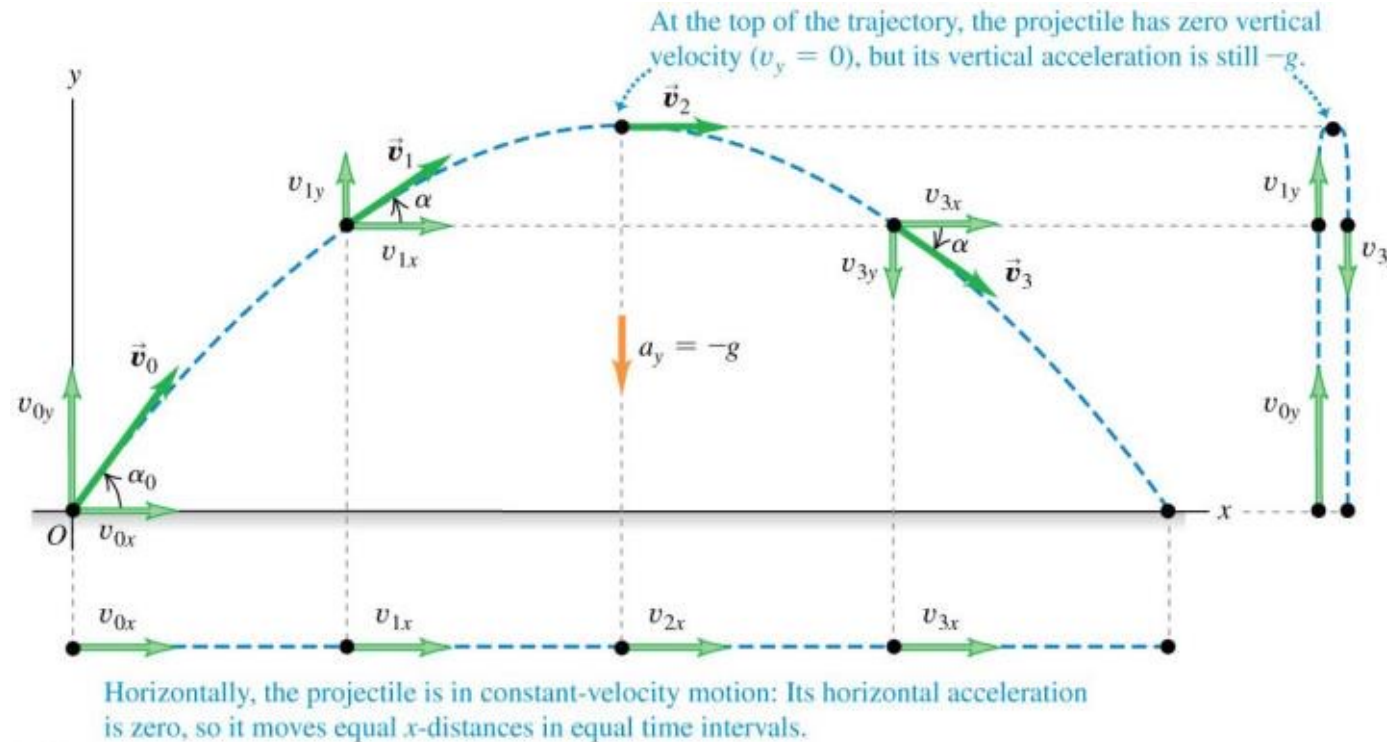
- A projectile moves in a vertical plane that contains the initial velocity vector $\vec{v}_0$.
- Its trajectory depends only on $\vec{v}_0$ and on the downward acceleration due to gravity.



All vectors can be decomposed into their x and y components.



- Acceleration is zero in x-direction, and constant in y-direction: $a_x = 0, a_y = -g$
- At the top of the trajectory: $v_y = 0$
- In the x-direction, where acceleration is 0, x-velocity v_x is constant.



Vertically, the projectile is in constant-acceleration motion in response to the earth's gravitational pull. Thus its vertical velocity *changes* by equal amounts during equal time intervals.

Horizontally, the projectile is in constant-velocity motion: Its horizontal acceleration is zero, so it moves equal x-distances in equal time intervals.

Projectile motion is parabolic (no air resistance)



Relative velocity

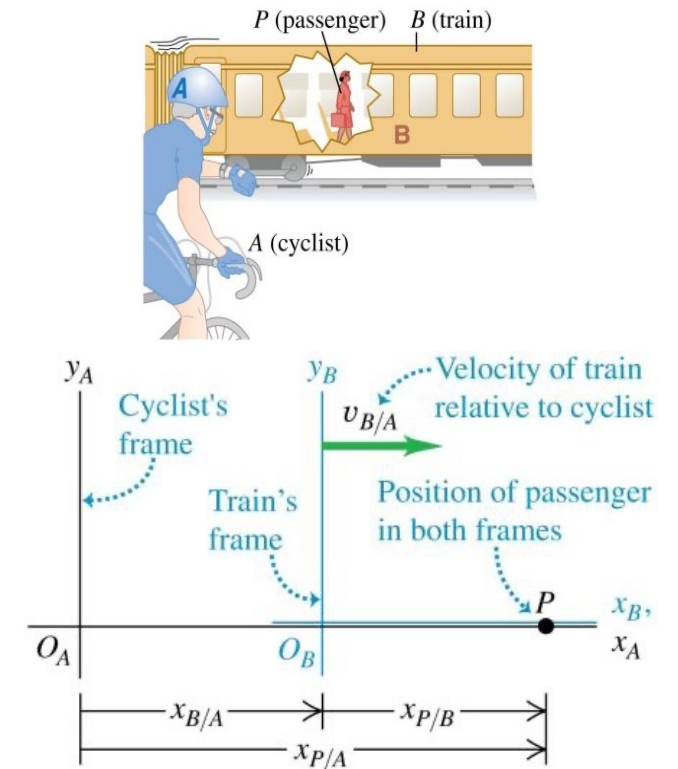
- The velocity of a moving body seen by a particular observer is called the velocity *relative* to that observer, or simply the **relative velocity**.
- A **frame of reference** is a coordinate system plus a time scale.
- If point P is moving relative to reference frame A, we denote the velocity of P relative to frame A as v_{PA} .
- If P is moving relative to frame B and frame B is moving relative to frame A, then the velocity of P relative to frame A is:

$$\vec{v}_{PA} = \vec{v}_{PB} + \vec{v}_{BA}$$

- Sometimes, it can be helpful to change the sign of the second term together with the subscripts (it doesn't change the equation):

$$\vec{v}_{PA} = \vec{v}_{PB} - \vec{v}_{AB}, \quad \because \vec{v}_{AB} = -\vec{v}_{BA}$$

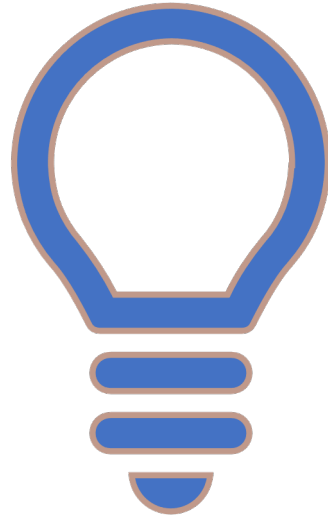
Visual and derivation



relative position: $x_{PA} = x_{PB} + x_{BA}$

differentiate wrt time, both sides

relative velocity: $v_{PA} = v_{PB} + v_{BA}$



Worked Examples

Example 1: Motion in 1D

On a moon landing, a rocket of ^m mass 47 000 kg is descending vertically at a steady speed of ^v 10 m s⁻¹. At a height of 80 m above the surface, a box dropped out of the rocket and landed on the surface of the moon. (Take g on moon as 1.6 m s^{-2} .)

Calculate

- the time taken for the toolbox to hit the surface after dropping out of the rocket, and
- the velocity when the box hit the surface.

$\uparrow +$
 $\downarrow$

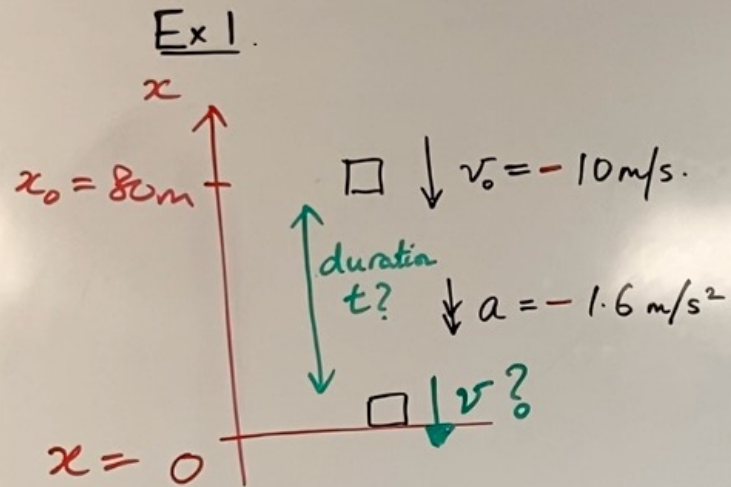
a) Using ~~st~~ $s = ut + \frac{1}{2}at^2$
 Under constant acceleration
 $s = 80, u = -10 \text{ m s}^{-1}, a = -1.6$
 $\frac{1}{2}(-1.6)t^2 + (-10)t - 80 = 0$
 $t = 5.54 \text{ or } -18.0$ (neg since we take time of drop to be pos)

b) $v^2 = u^2 + 2as$
 $= (-10)^2 + 2(-1.6)(-80)$
 $v^2 = 356$
 $v = -18.87 \text{ or } 18.87$
 $\therefore v = -18.87 \text{ m/s}$ (neg since moving down)

Answers:

- $t = 5.54 \text{ s}$
- $v = -18.87 \text{ m/s}$

Solution to Example 1: Motion in 1D



Eg of motion (const acc.)

- ① $v = v_0 + at$
- ② $\underbrace{x - x_0}_s = v_0 t + \frac{1}{2}at^2$
- ③ $v^2 = v_0^2 + 2a(\underbrace{x - x_0}_s)$

a) Find t

$$0 - 80 = -10t + \frac{1}{2}(-1.6)t^2$$

$\therefore$

$$t = \begin{cases} -18.04\text{s} & \text{reject} \\ \text{OR} \\ 5.54\text{s} & \checkmark \end{cases}$$

b) Find v

Use ①:

$$v = -10 + (-1.6)(5.54) \\ = -18.9\text{m/s}$$

Use ③:

$$v^2 = (-10)^2 + 2(-1.6)(80)$$

$$v = \pm 18.9\text{m/s}$$

$\uparrow$
reject positive value.

Example 2: Motion in 1D

(Modified) Giancoli pg 36 Example 2-16

A person throws a ball upward into the air with an initial velocity of 15 m s^{-1} .

Take $g = 9.8 \text{ m s}^{-2}$. Ignore air resistance.

- What is the acceleration at the highest point (B)?
- Sketch the velocity-time, acceleration-time and the displacement-time graph of the ball from point A to C.
- Calculate the maximum height of the ball.
- Calculate how long the ball comes back to the hand.
- Calculate at what time the ball passes a point 8.00 m above the person's hand.

c) using $\vec{v} = \vec{u} + \vec{a}t$

$$s = \frac{v^2 - u^2}{2a} = \frac{(0)^2 - (15)^2}{2(-9.8)} = 11.47$$

Displacement is 11.5 m

d) $s = ut + \frac{1}{2}at^2$

$(0) = (-15)(t) + \frac{1}{2}(-9.8)(t^2)$

Solve for t , $t = 0$ 3.06 $\therefore t = 3.06 \text{ s}$

crept since refers to the hand

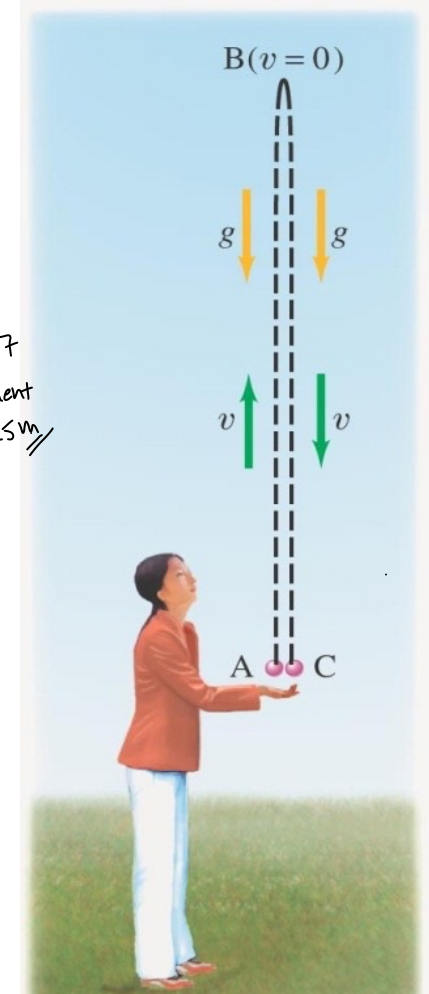
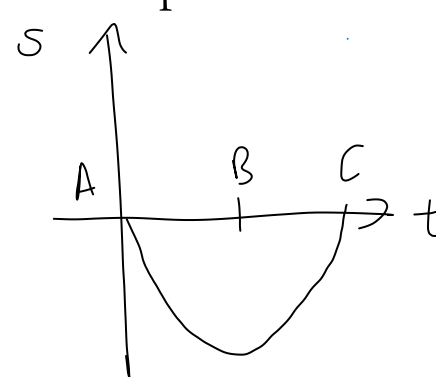
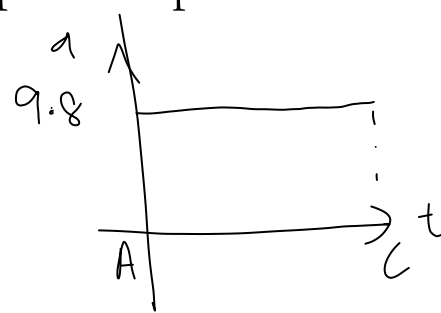
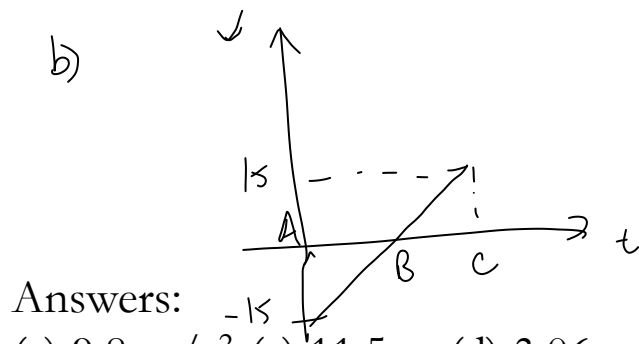


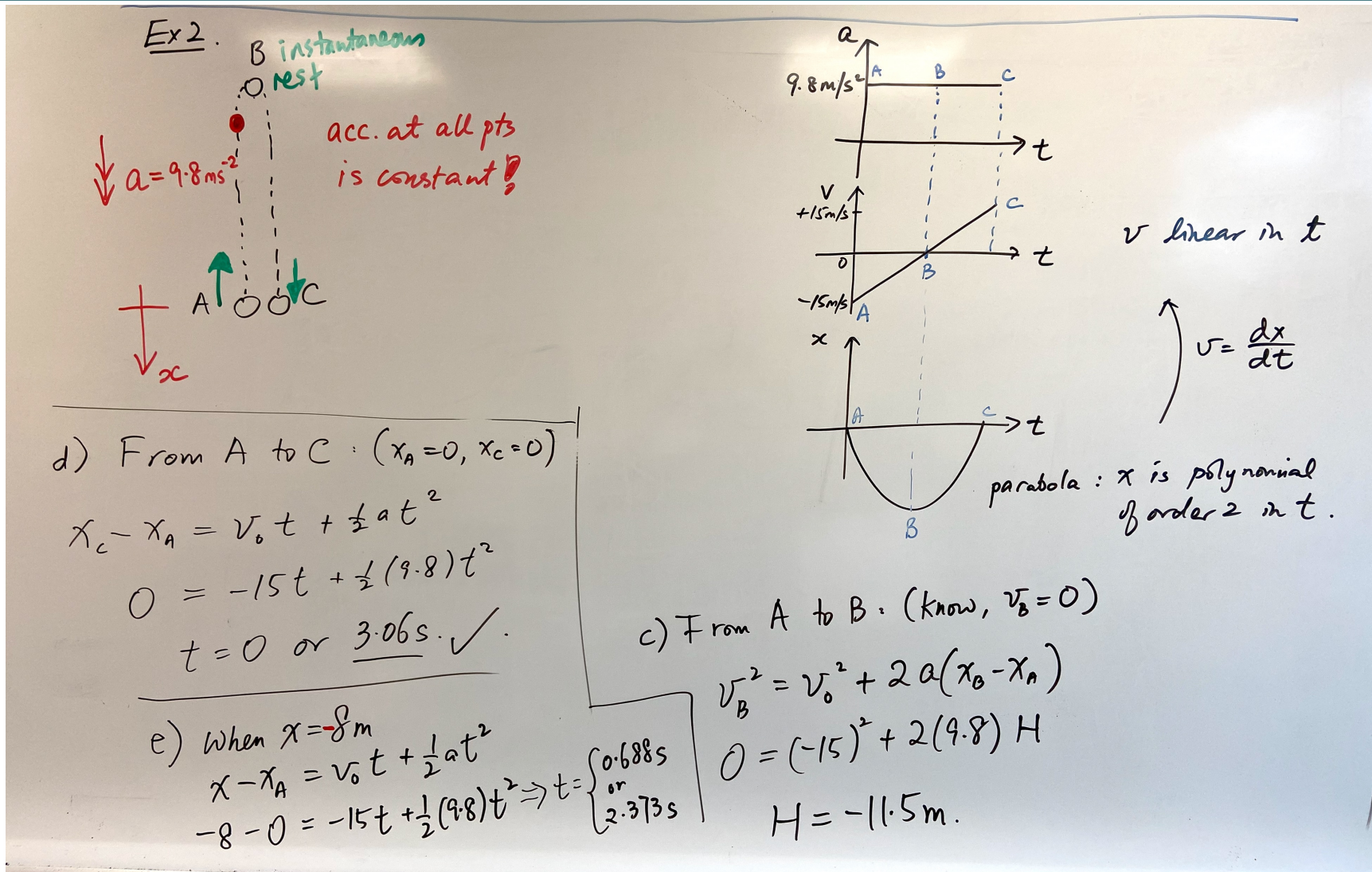
Image source: Giancoli



Answers:

(a) 9.8 m/s^2 (c) 11.5 m (d) 3.06 s (e) $0.69 \text{ s}, 2.37 \text{ s}$

Solution to Example 2: Motion in 1D



Example 3: Motion in 2D (Projectile motion)

Giancoli, Example 3-6

A movie stunt driver on a motorcycle speeds horizontally off a 50.0 m high cliff. Ignore air resistance.

- How fast must the motorcycle leave the cliff top to land on level ground below, 90.0 m from the base of the cliff where the cameras are?
- What is the velocity of the motorcycle just before it hits the ground below?

a) Account for y-component

$$s = ut + \frac{1}{2}at^2$$

$$(-50) = 0 + \frac{1}{2}(-9.8)t^2$$

$$t = \pm 3.19 \text{ s (rej. -ve)}$$

$$\therefore t = 3.19 \text{ s}$$

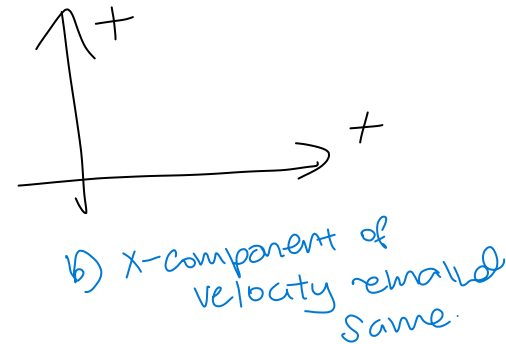
Account of x-component

$$s = ut + \frac{1}{2}at^2$$

$$a = 0 \quad u = \frac{s}{t} = \frac{90}{3.19} = 28.2 \text{ m/s}$$

Answers:

(a) 28.2 m/s (b) 42.1 m/s at angle 48° below horizontal



$$v_y = u_y + at$$

$$= 0 + (-9.8)(3.19)$$

$$= -31.262$$

$$|v| = \sqrt{28.2^2 + 31.262^2} = 42.1 \text{ m/s}$$

$$\theta = \tan^{-1} \frac{31.262}{28.2} = 48^\circ$$

48° below horizontal.

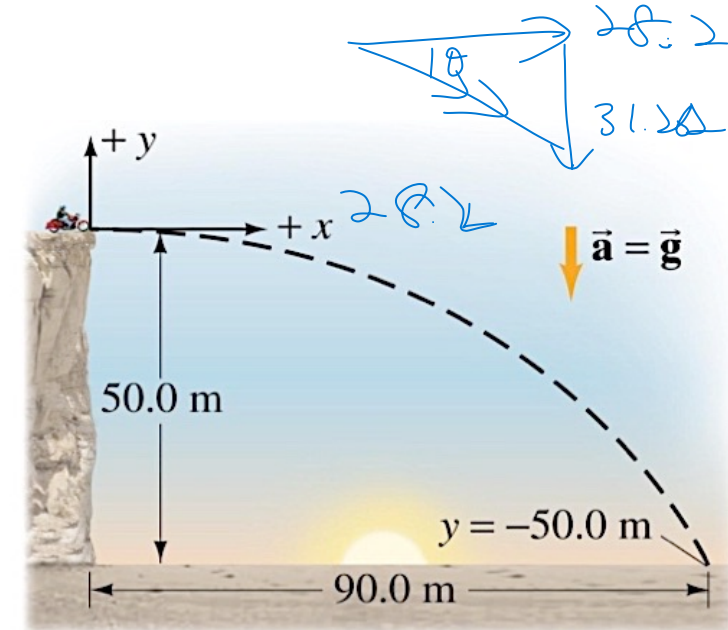
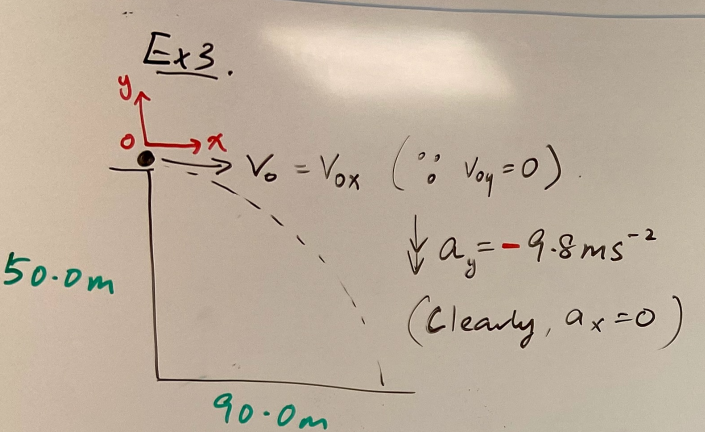


Image source: Giancoli

Solution to Example 3: Motion in 2D (Projectile motion)

Ex3.



$v_0 = v_{0x}$ ($\because v_{0y} = 0$)
 $a_y = -9.8 \text{ ms}^{-2}$
(Clearly, $a_x = 0$)

50.0m
90.0m

$v_x = v_{0x}$
 $v_y = v_{0y} + a_y t = (9.8 \times 3.19) \text{ m/s}$
 $v = \sqrt{v_y^2 + v_x^2} = 42.08 \text{ m/s}$
 $\tan \theta = \frac{v_y}{v_x} \Rightarrow \theta = 48.0^\circ$

x-dir: $\Delta x = v_{0x} t$
 $v_{0x} = \frac{\Delta x}{t} \quad \text{--- (1)}$

y-dir: $\Delta y = v_{0y} t + \frac{1}{2} a_y t^2$
 $-50.0 - 0 = \frac{1}{2} (-9.8) t^2$
 $t^2 = \frac{2\Delta y}{a} \quad \text{--- (2)}$
 $t = 3.19 \text{ s}$

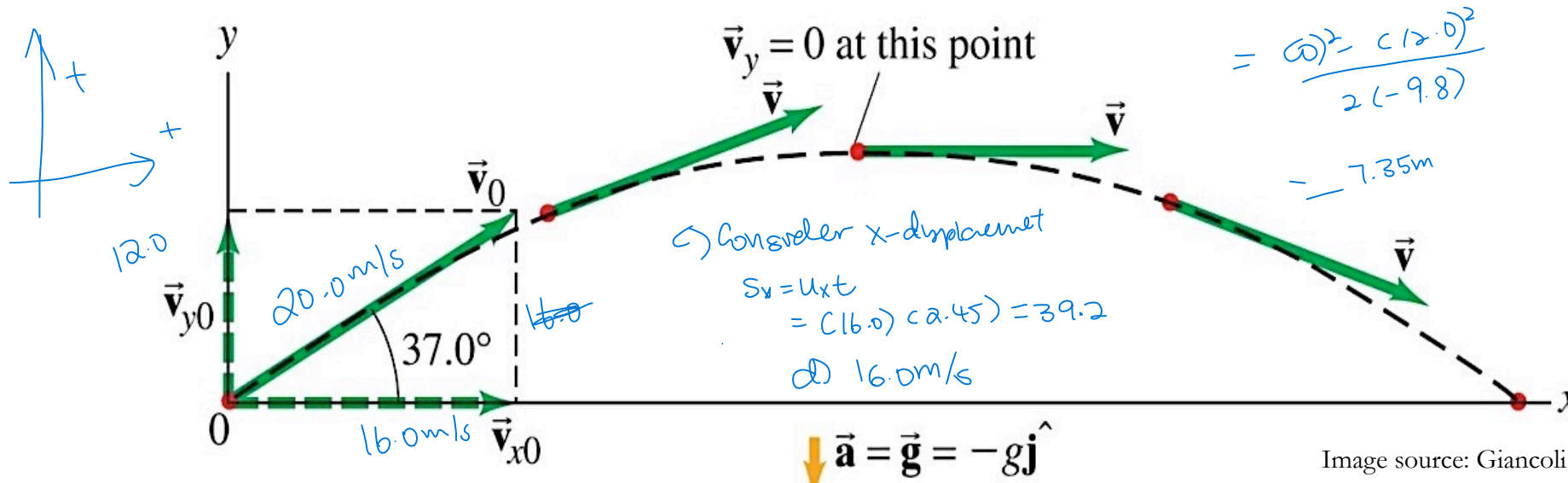
$v_{0x} = \frac{\Delta x}{\sqrt{2\Delta y/a}} = 28.2 \text{ m/s}$
 $= 28.2 \times 10^{-3} \text{ km/s}$
 $= 28.2 \times 10^{-3} \times 60 \times 60 \text{ km/h}$
 $\approx 101 \text{ km/h}$

Example 4: Motion in 2D (Projectile motion)

Giancoli, Example 3-7

A football is kicked at an angle $\theta_0 = 37.0^\circ$ with a velocity of 20.0 m/s, as shown. Assume the ball leaves the foot at ground level, and ignore air resistance and rotation of the ball. Calculate

- the maximum height
- the time of travel before the football hits the ground,
- how far away it hits the ground,
- the velocity vector at the maximum height, and
- the acceleration vector at maximum height.



Answers:

- 7.35 m
- 2.45 s
- 39.2 m
- 16.0 m/s
- 9.8 m/s² downwards

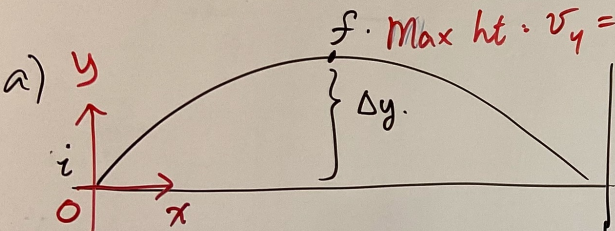
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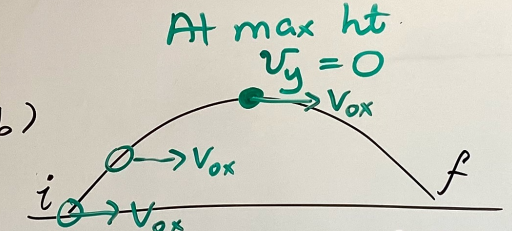
Solution to Example 4: Motion in 2D (Projectile motion)

Ex 4.

$V_{0y} = V_0 \sin 37^\circ = 12.0 \text{ m/s}$
 $V_0 = 20.0 \text{ m/s}$
 $V_{0x} = V_0 \cos 37^\circ = 16.0 \text{ m/s}$

$\downarrow a_y = -9.8 \text{ m/s}^2$

a) 
 f. Max ht. $v_y = 0$
 $\cancel{v_y}^2 = v_{0y}^2 + 2a_y \Delta y$
 $0 = 12.0^2 + 2(-9.8)\Delta y$
 Max ht, $\Delta y = 7.35 \text{ m}$

b) 
 At max ht, $v_y = 0$
 $\Delta y = v_{0y}t + \frac{1}{2}a_y t^2$
 $0 = 12.0t + \frac{1}{2}(-9.8)t^2$
 $t = 0 \text{ s or } 2.45 \text{ s}$
 answer

c) $\Delta x = v_{0x}t = (16.0 \cdot 2.45) \text{ m} = 39.2 \text{ m}$

d) At max ht,
 $v_y = 0$
 $v_x = v_{0x}$
 $\therefore V = v_{0x} = 16.0 \text{ m/s}$

e) $\vec{a}$ const
 $a = -9.8 \text{ m/s}^2$

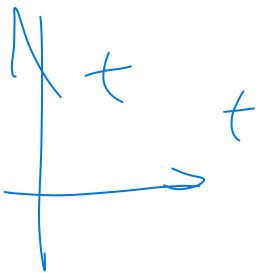
Example 5: Relative velocity

Giancoli, Ch 3, Problem 58

A man walks at a speed of 0.70 m s^{-1} across his raft relative to the raft. The raft is drifting along the river at a speed of 1.50 m s^{-1} relative to the shore.

$M \rightarrow$ Man $R \rightarrow$ Raft
 $S \rightarrow$ shore.

What is the man's velocity relative to the shore?



$$V_{M/Ry} = 0.7 \text{ m/s}$$

$$V_{M/Rx} = 0 \text{ m/s}$$

$$V_{R/Sx} = 1.50 \text{ m/s}$$

$$V_{R/Sy} = 0 \text{ m/s}$$

$$V_{Msy} = V_{M/Ry} + V_{R/Sy} = 0.7 \text{ m/s} + 0$$

$$V_{Msx} = V_{M/Rx} + V_{R/Sx} = 0 + 1.5$$

$$|V_{Ms}| = 1.66 \quad \theta = \tan^{-1} \frac{0.7}{1.5} = 25.0^\circ \text{ above horizontal}$$

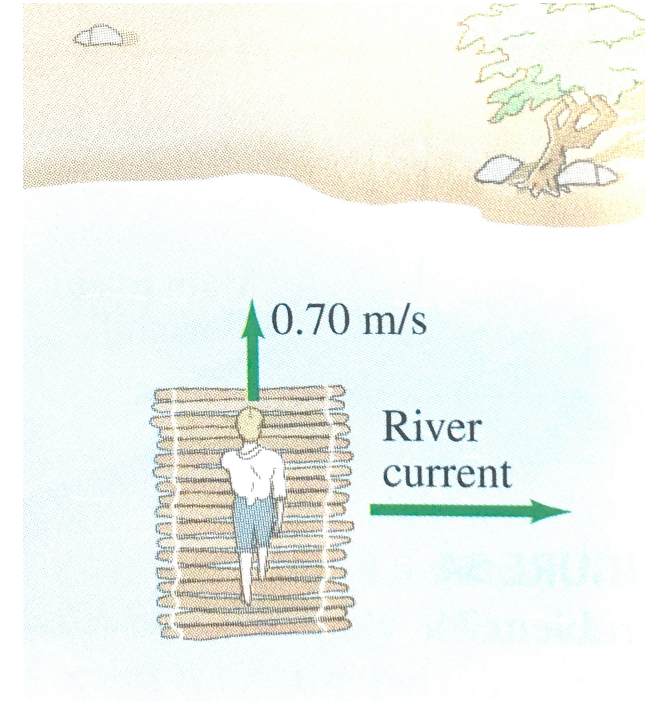


Image source: Giancoli

Solution to Example 5: Relative velocity

Ex 5 Relative velocity

shore

$\vec{V}_{rs} = \begin{pmatrix} 1.5 \\ 0 \end{pmatrix} \text{ m/s}$

$\vec{V}_{mr} = \begin{pmatrix} 0 \\ 0.7 \end{pmatrix} \text{ m/s}$

$\vec{V}_{ms} = \vec{V}_{mr} + \vec{V}_{rs}$

$\vec{V}_{ms} = \begin{pmatrix} 1.5 \\ 0.7 \end{pmatrix} \text{ m/s}$

$|\vec{V}_{ms}| = \sqrt{1.5^2 + 0.7^2} = 1.66 \text{ m/s}$

$\theta = \tan^{-1}\left(\frac{1.5}{0.7}\right) = 65^\circ$

The diagram shows a coordinate system with a vertical y-axis and a horizontal x-axis. A horizontal line represents the 'shore'. Below it, a box labeled 'raft' contains a stick figure labeled 'man'. A red arrow labeled $\vec{V}_{mr}$ points upwards from the man. To the right of the raft, a red arrow labeled $\vec{V}_{rs}$ points to the right. Below the raft, a vector diagram shows $\vec{V}_{mr}$ as a vertical vector, $\vec{V}_{rs}$ as a horizontal vector, and their resultant $\vec{V}_{ms}$ as the hypotenuse of a right-angled triangle. The angle θ is marked between $\vec{V}_{mr}$ and $\vec{V}_{ms}$.

Further reading and watching

- Young and Freedman, Chapters 1, 2, 3
- MasteringPhysics Dynamic Study Module : Kinematics

Free fall

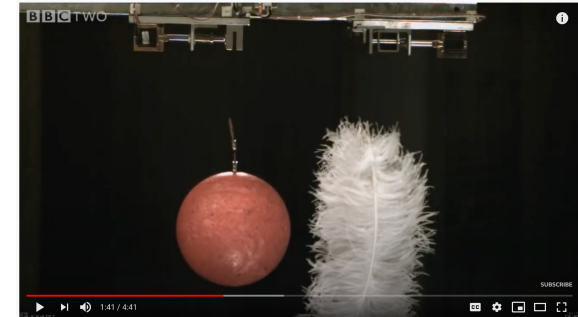
- Brian Cox visits the world's biggest vacuum:
<https://youtu.be/E43-CfukEgs>

Redefinition of the kilogram

- The kg is dead, long live the kg.
https://youtu.be/c_e1wITe_ig [Veritasium]
- Wikipedia: Planck's constant
https://en.wikipedia.org/wiki/Planck_constant
(Go to section “Understanding the fixing of the value of h ”)

Orthogonal directions are independent (projectile motion)

- Paul Hewitt: <https://youtu.be/qErh402eJgI>



BBC: #HumanUniverse
Brian Cox visits the world's biggest vacuum | Human Universe - BBC



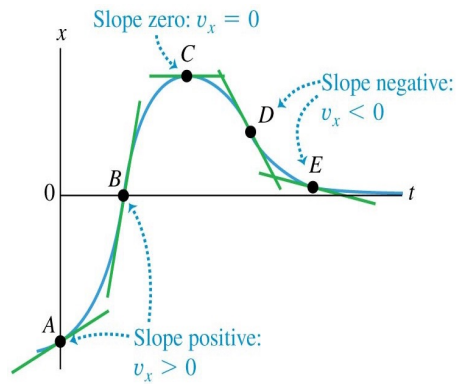
The kg is dead, long live the kg



conceptual physics Projectile Motion

Summary: General relations in Kinematics

Gradient of a position-time graph is velocity.

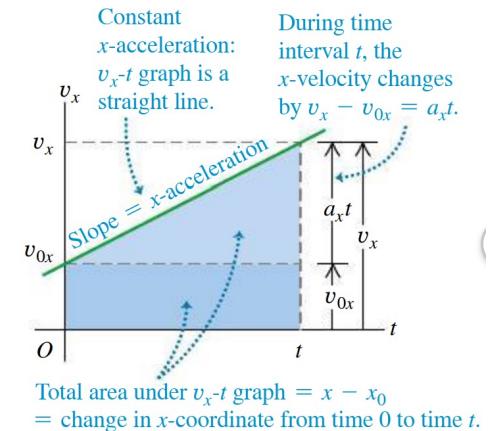


$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{x}}{\Delta t} = \frac{d\vec{x}}{dt}$$

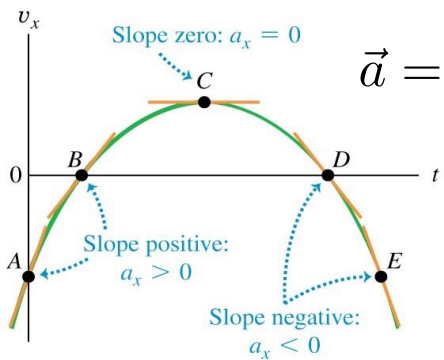
$$\Rightarrow \int_{\vec{x}_i}^{\vec{x}_f} d\vec{x} = \int_{t_i}^{t_f} \vec{v}(t) dt$$

$$\vec{x}_f - \vec{x}_i = \int_{t_i}^{t_f} \vec{v}(t) dt$$

Area under a velocity-time graph is displacement (change in position).



Gradient of a velocity-time graph is acceleration.

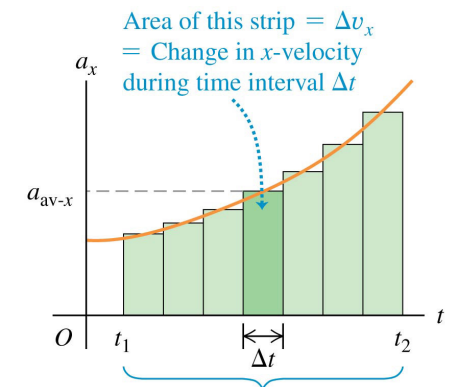


$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2 \vec{x}}{dt^2}$$

$$\Rightarrow \int_{\vec{v}_i}^{\vec{v}_f} d\vec{v} = \int_{t_i}^{t_f} \vec{a}(t) dt$$

$$\vec{v}_f - \vec{v}_i = \int_{t_i}^{t_f} \vec{a}(t) dt$$

Area under an acceleration-time graph is change in velocity.



Summary: Equations of motion (const acc), Relative velocity

Equations of motion (const acc)

$$v(t) = v_0 + at$$

$$x(t) = x_0 + v_0t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

Relative velocity

$$\vec{v}_{PA} = \vec{v}_{PB} + \vec{v}_{BA}$$